Unit <IV>

Flip-flop

sequential circuits :-

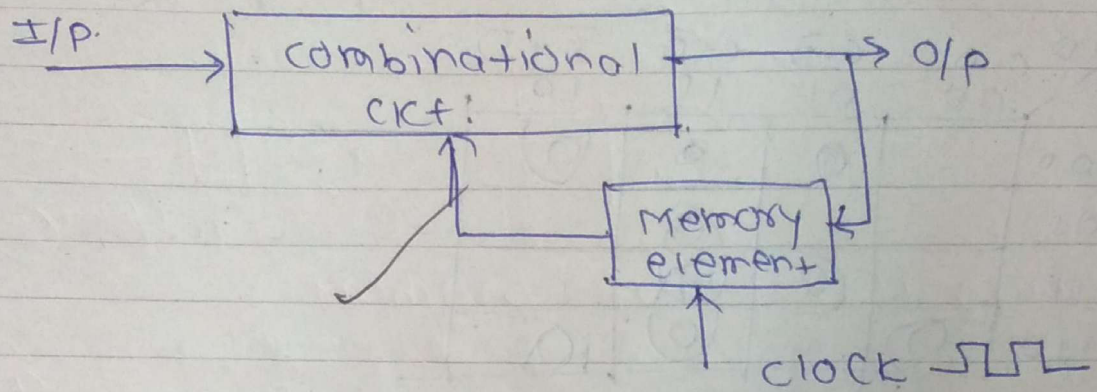


fig :- Block diagram of a sequential ckt.

1) The memory element is called F/F.
2) F/F is basic memory element.

$f = 1/T$

clock signal :-

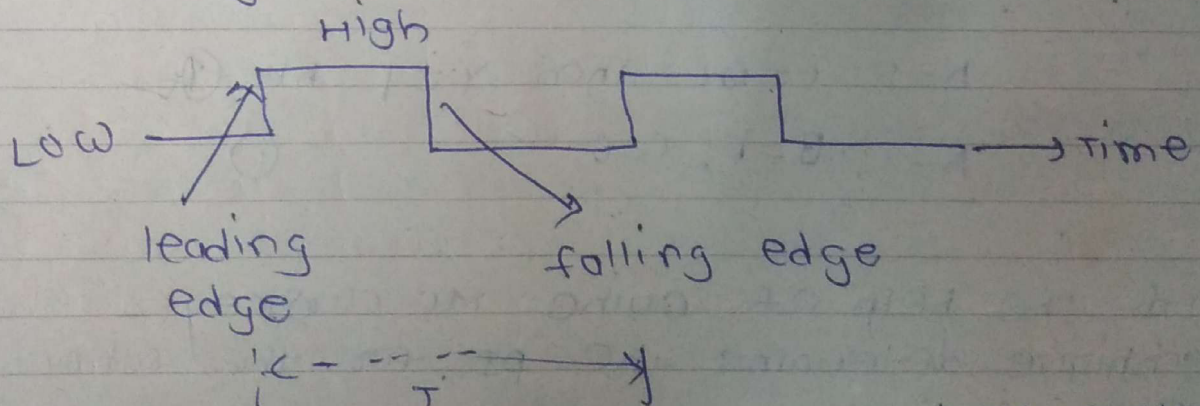
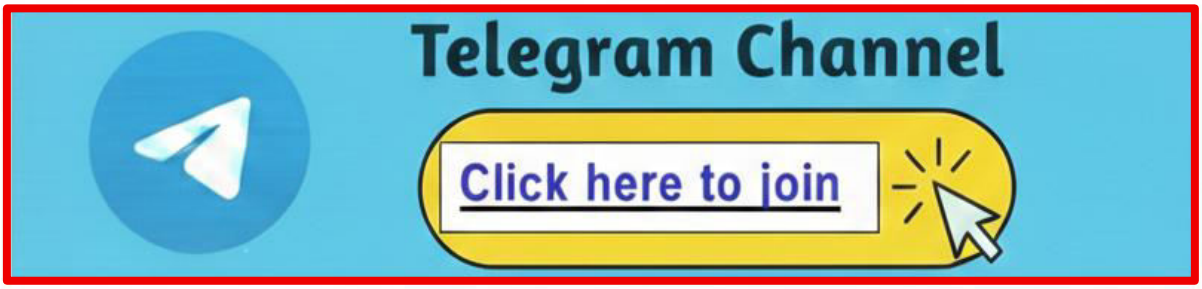


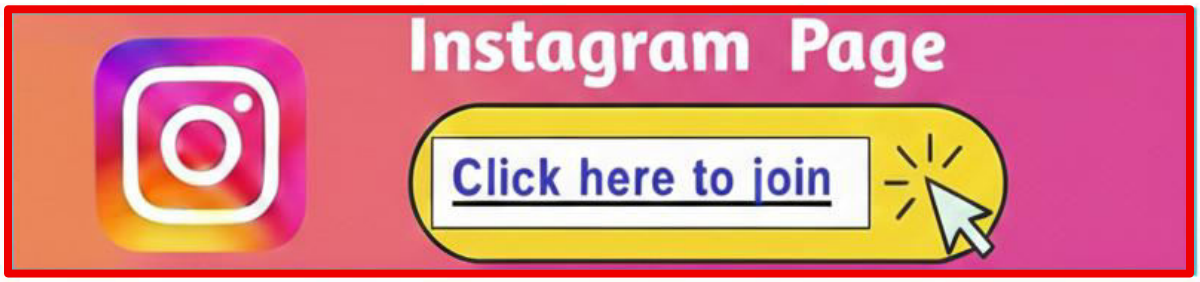
fig :- clock signal

comparision of combinational & sequential ckt :-

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Sr. No.	parameters	combinational circuits.	sequential circuits.
1)	output depends on	I/p present at that instant of time	present i/p & past i/p / o/p.
2)	Memory	Not necessary	Necessary
3)	clock i/p:	Not necessary	Necessary.
4)	e.g -:	Adder, substractor, code converters.	F/F, counter, shift register.

* 1-bit memory cell (Bistable element)-:
  - Two stable state
    - logic '0'
    - logic '1'

  use NAND OR NOR gate

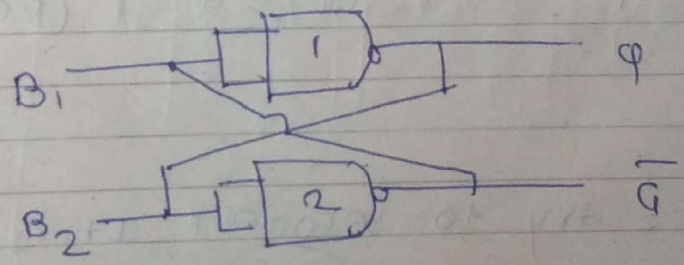


fig -: A cross coupled inverter as memory element.

operation -:

$Q = 1$ then $\overline{Q} = 0$.

$Q = 1$ $\overline{Q} = 0$ → set state (1 state)

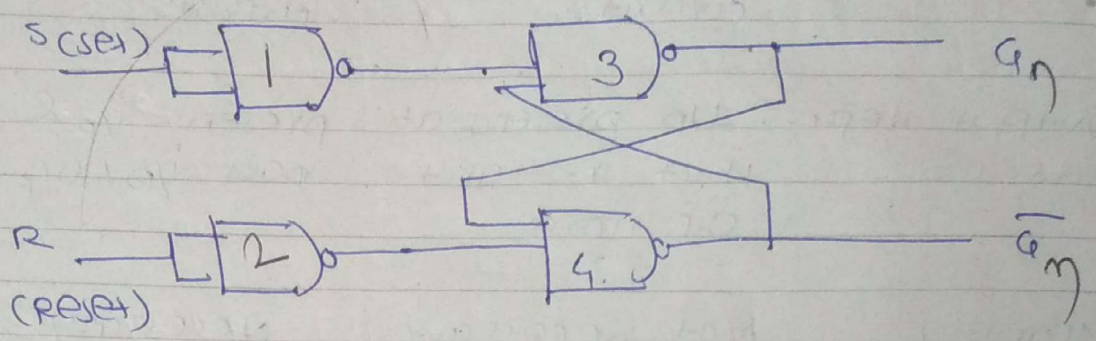
$Q = 0$ $\overline{Q} = 1$ → reset state (0 state)

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store's '1' bit information.

Latch :-



Modified memory cell (Latch)

① S=R=0 (NC in o/p) (race)

Let,

Q=0, $\bar{Q}$=1 Q_n

② S=1, R=0 (set)

S=1, R=0, Q=1 & $\bar{Q}$=0 (set)

③ S=0 R=1

S=0, R=1 Q=0 & $\bar{Q}$=1 (Reset)/(reset)

④ S=R=1

Q & $\bar{Q}$ = try to become 1, It is not allowed. (race)

(N.C.)

* Clocked S-R F/F :-

Ck = 1 operation same as F/F.

Ck = 0 o/p are 1

where clk is present then the ckt respond to i/p.

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I/P

Sn Rn

0 0

0 1

1 0

1 1

O/P

Q_{n+1} $\overline{Q_{n+1}}$

Q_n $\overline{Q_n}$

0 1

1 0

Race (?) Race (?)

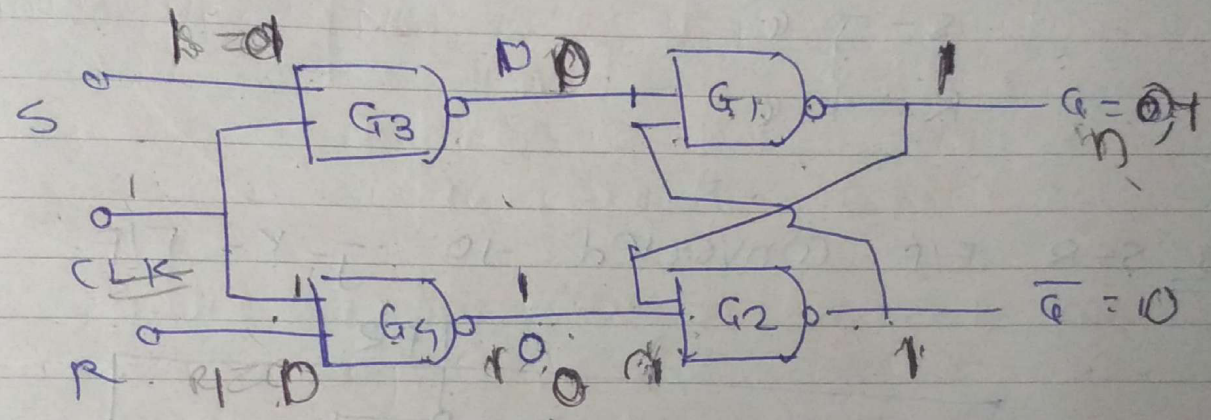


Fig :- Clocked S/R F/F.

Case I CLK = 0 operation inactive
O/p will be forced to 1.

Case II CLK = 1

1) S = R = 0 → N.C.

2) S = 0, R = 1 →

$\overline{Q} = 1$ $Q = 0$ → Reset condn.

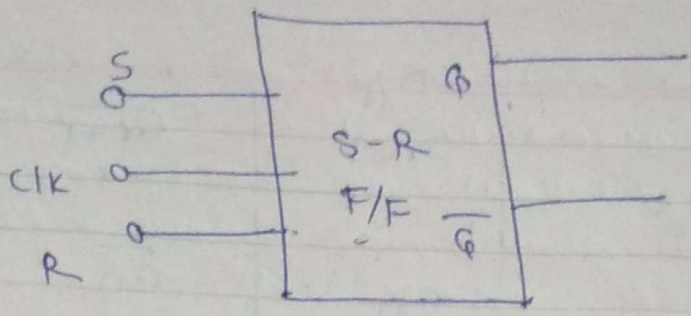
3) S = 1, R = 0

$Q = 1$, $\overline{Q} = 0$ → set condn.

4) S = R = 1 → Race Ground condition

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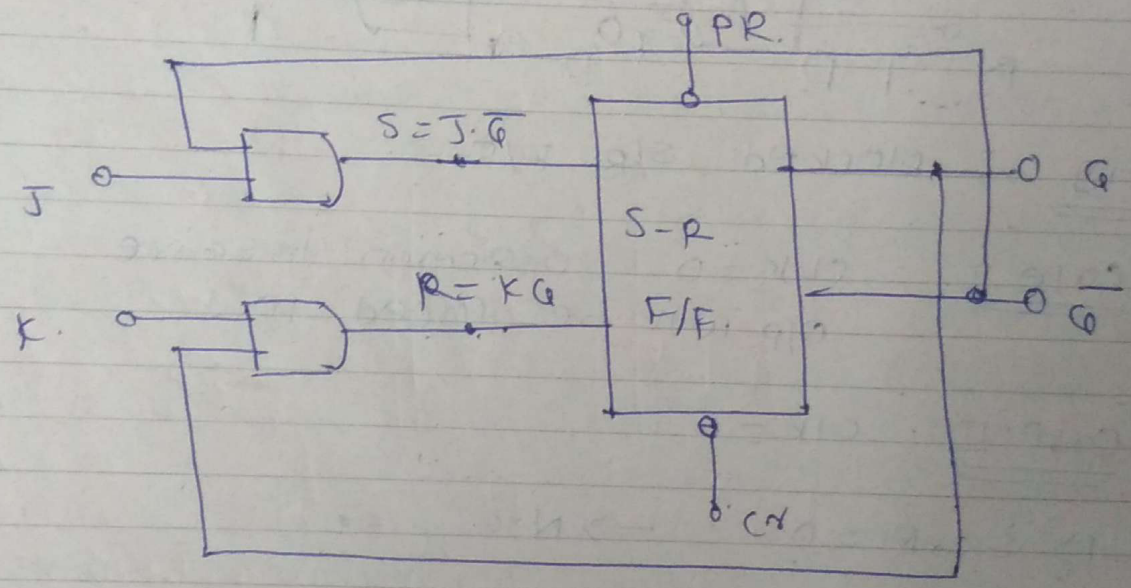
(1) –

(2) J-K – F/F – :

$S = J \cdot \bar{Q}$

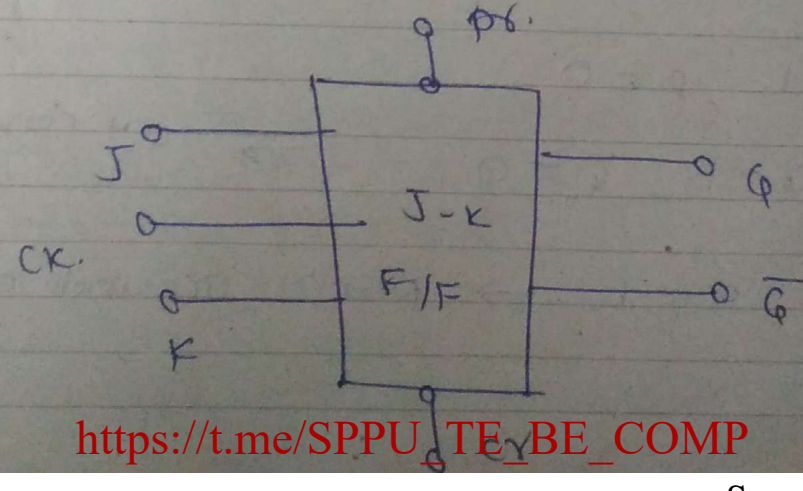
$R = K Q$

fig S-R F/F converted to J-K- F/F.



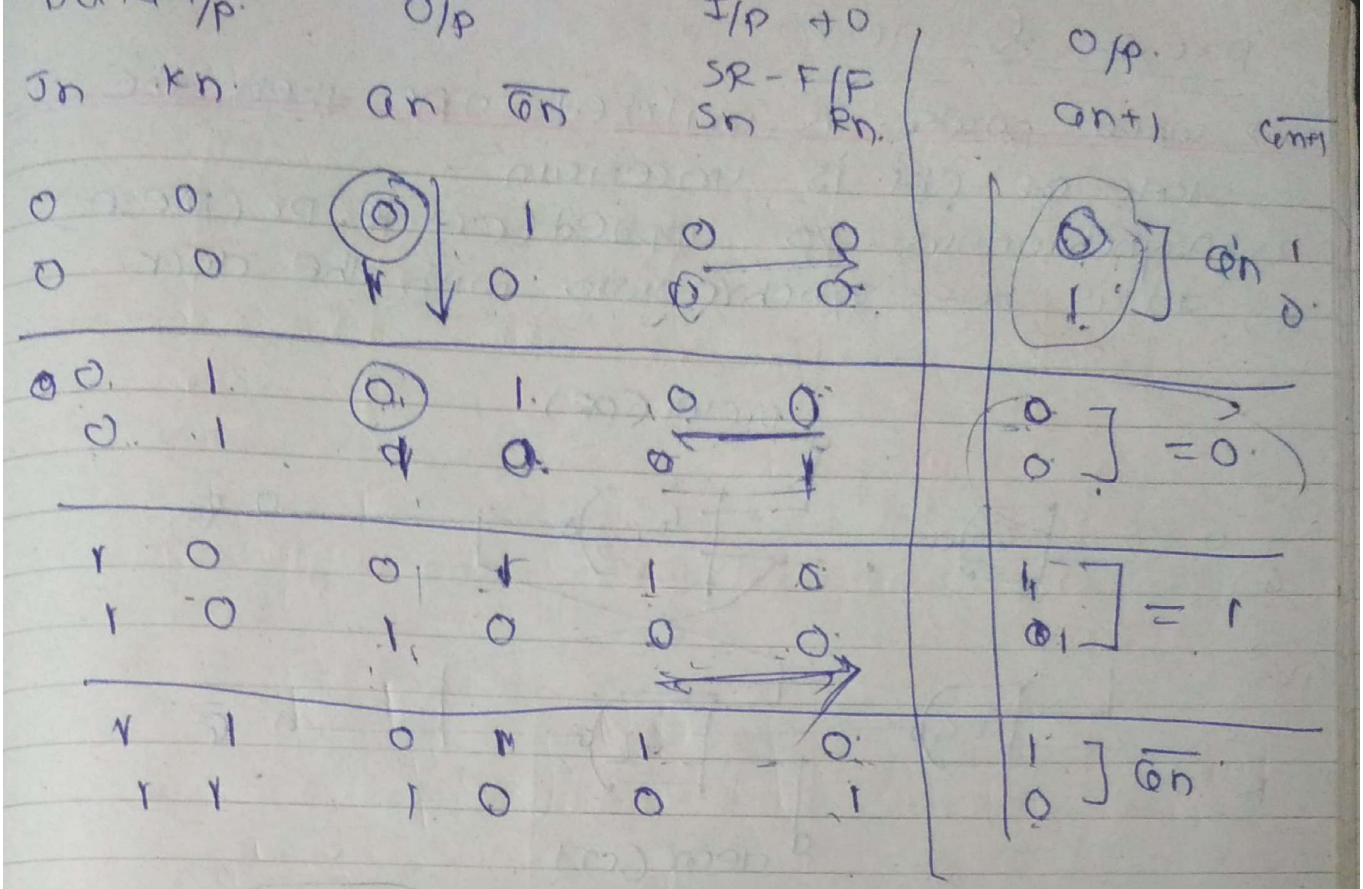
for the condition S=R=1

Logic symbol of J-K- F/F – :



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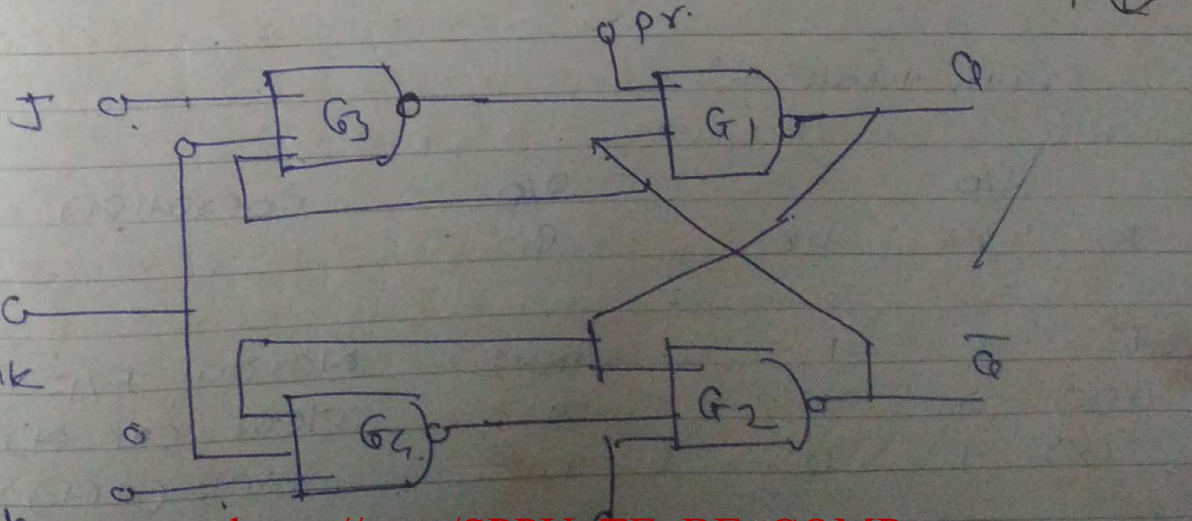
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i/p = 00 means take previous condn.

Jn	Kn	Qn+1	$\overline{Q_{n+1}}$	
0	0	Qn	$\overline{Q_n}$	No change
0	1	0	1	Reset
1	0	1	0	Set
1	1	$\overline{Q_n}$	Qn	Toggle

Truth table for J-K F/F.

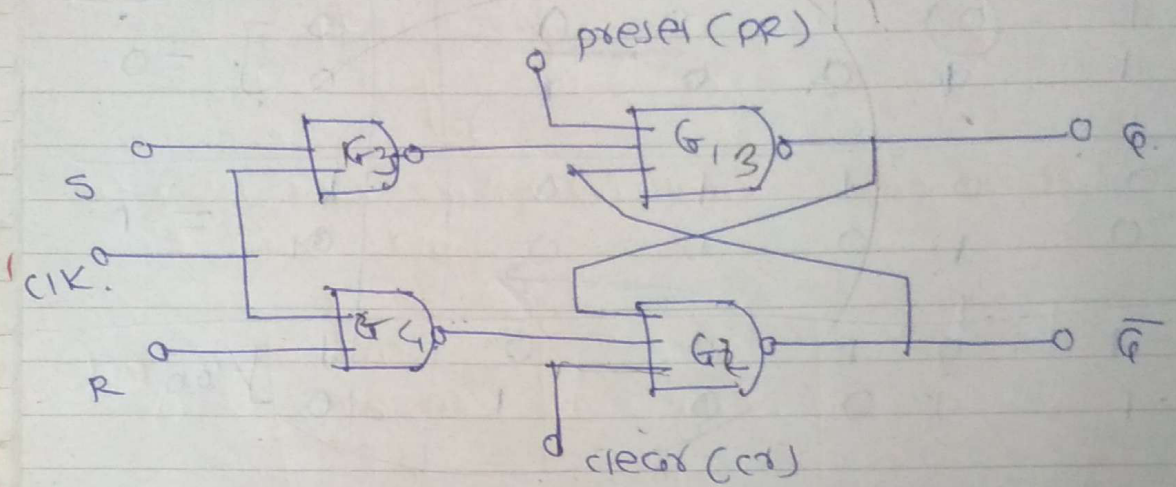


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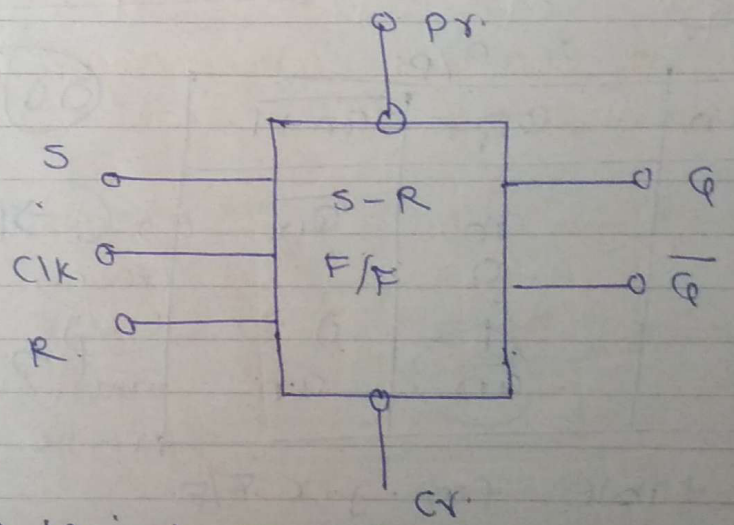
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preset & clear :-

1) When power is switched off, then the state of ckt. is uncertain
2) Asynchronous i/p applied (cr) & pr (preset)
3) It is not synchronism with the clock.



(a) An S-R F/F with preset & clear.



(b) logical symbol.

Truth table :-

I/p			O/p	operation
CK	cr	pr	Q	
1	1	1	Q_{n+1}	Normal F/F
0(x)	0	1	0	clear (reset)
0(x)	1	0	1	preset (set)

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① $\overline{PR}=0$ $\overline{CR}=1$ then sets the F/F. $\overline{PR}=0$

② $\overline{PR}=1$ & $\overline{CR}=0$ then F/F is reset BC

$\overline{PR}=0$ & $\overline{CR}=0$ is not used, since this leads to uncertain state.

③ D-type F/F :-

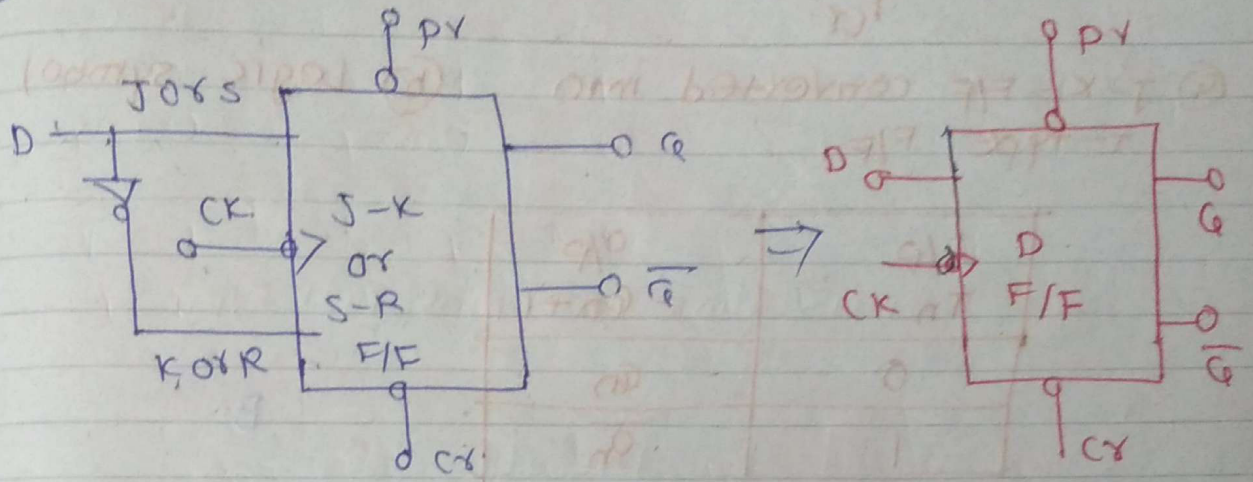


Fig :- J-K OR S-R F/F is converted to D F/F

logical symbol of D F/F.

Truth table

I/p	O/p
D_n	Q_{n+1}
0	0
1	1

1) D = Delay F/F.
2) It is used to store 1-bit information.

④ T-type F/F :-

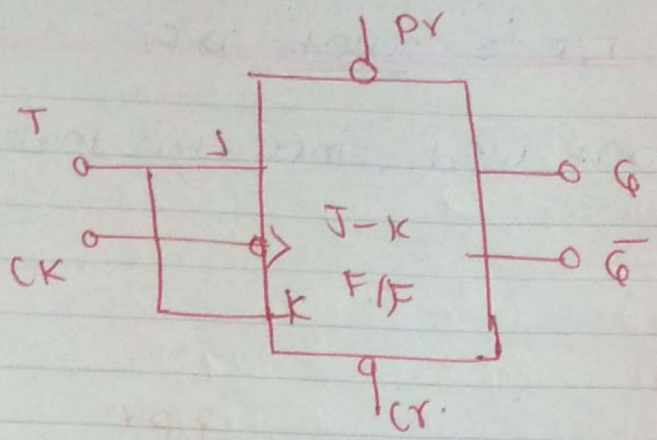
In J-K F/F J=K the resulting F/F is called Toggle F/F or T F/F.

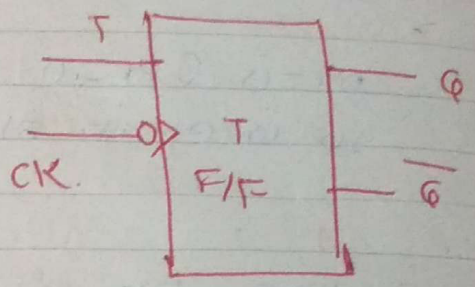
T=1 act as toggle s/w

For every clock pulse the o/p will changes.

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(a) J-K F/F converted into T-type F/F.

(b) Logic symbol

I/P T_n	O/P Q_{n+1}
0	Q_n
1	$\bar{Q}_n$

B

S=R=1 not allowed acts as toggle s/w

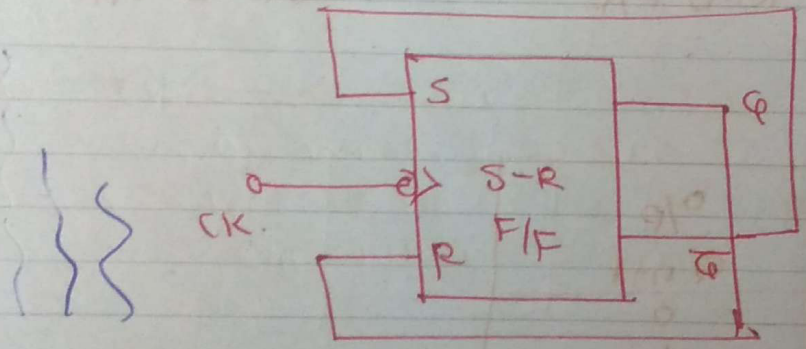


fig :- S-R F/F act as toggle s/w.

* The Master-Slave J-K-F/F :-

1) slave always follow master
2) Clk = 0 Master - inactive
   Slave - active
3) Clk = 1 Master - active
   Slave - inactive

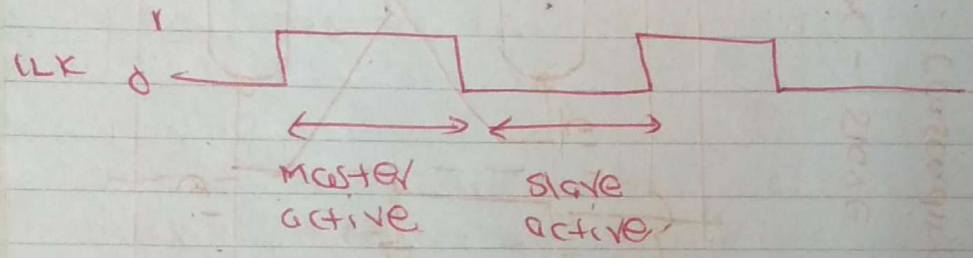
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Truth table :-

I/p			O/p		Remark
Clk	J	K	Q_{n+1}	$\overline{Q_{n+1}}$	
X	0	0	Q_n	$\overline{Q_n}$	NC
↓ (1)	0	0	Q_n	$\overline{Q_n}$	NC
↓ (1)	0	1	0	1	Reset
↓ (1)	1	0	1	0	set
↓ (1)	1	1	$\overline{Q_n}$	Q_n	Toggle

fig :- Truth table of M-S (J-K F/F).



I) clock = X , J = K = 0 Clk = 1
O/p of master i.e. Q_1 & $\overline{Q_1}$ will not change.
S-R i/p to the slave remains unchanged
O/p will not changes.

II) same as 1st

III) clock = 1 J = 0 & K = 1
Clk = 1 Master active, slave inactive
J = 0, K = 1, $Q_1 = 0$ & $\overline{Q} = 1$ i.e. S = 0 R = 1
Clk = 0 Master inactive slave active
$\overline{Q} = 1$ $Q_1 = 0$.
O/p of slave $Q = 0$ $\overline{Q} = 1$
we get stable o/p from the master-slave

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IV) clk = ↓, J=1, K=0

clk = 1, Master active slave inactive

$Q_1 = 1$ $\bar{Q}_1 = 0$

clk = 0, Master inactive, slave active

$Q_1 = 1$ $\bar{Q}_1 = 0$

V) clk = 1 J = K = 1

clk = 1, Master active, slave inactive

∴ o/p of master will toggle so se also will be inverted

clk = 0

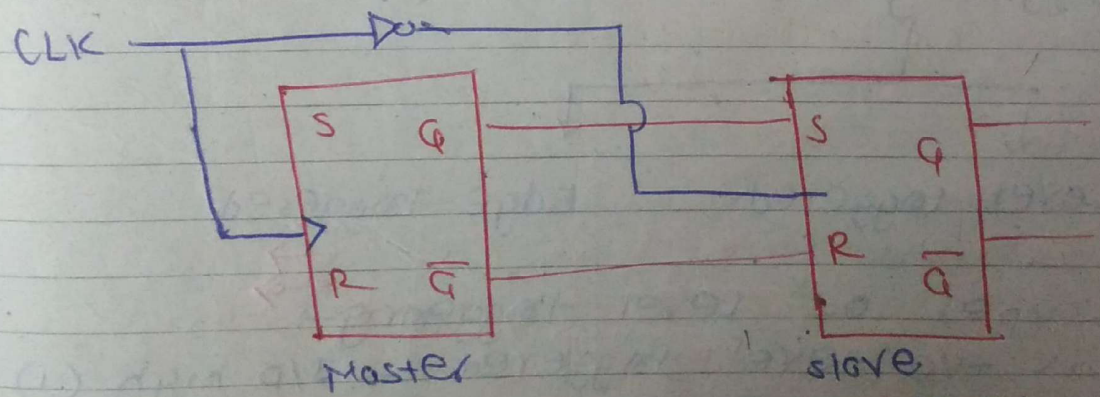
o/p of slave will toggle

To avoid the multiple toggeling condition which leads race around condition. Thus the m/s will respond (avoid) race around condn.

Advantage :-

1) slave also follow master
2) Multiple toggeling or race around condition is avoided.

Master slave S-R F/F



+ve level trigger master

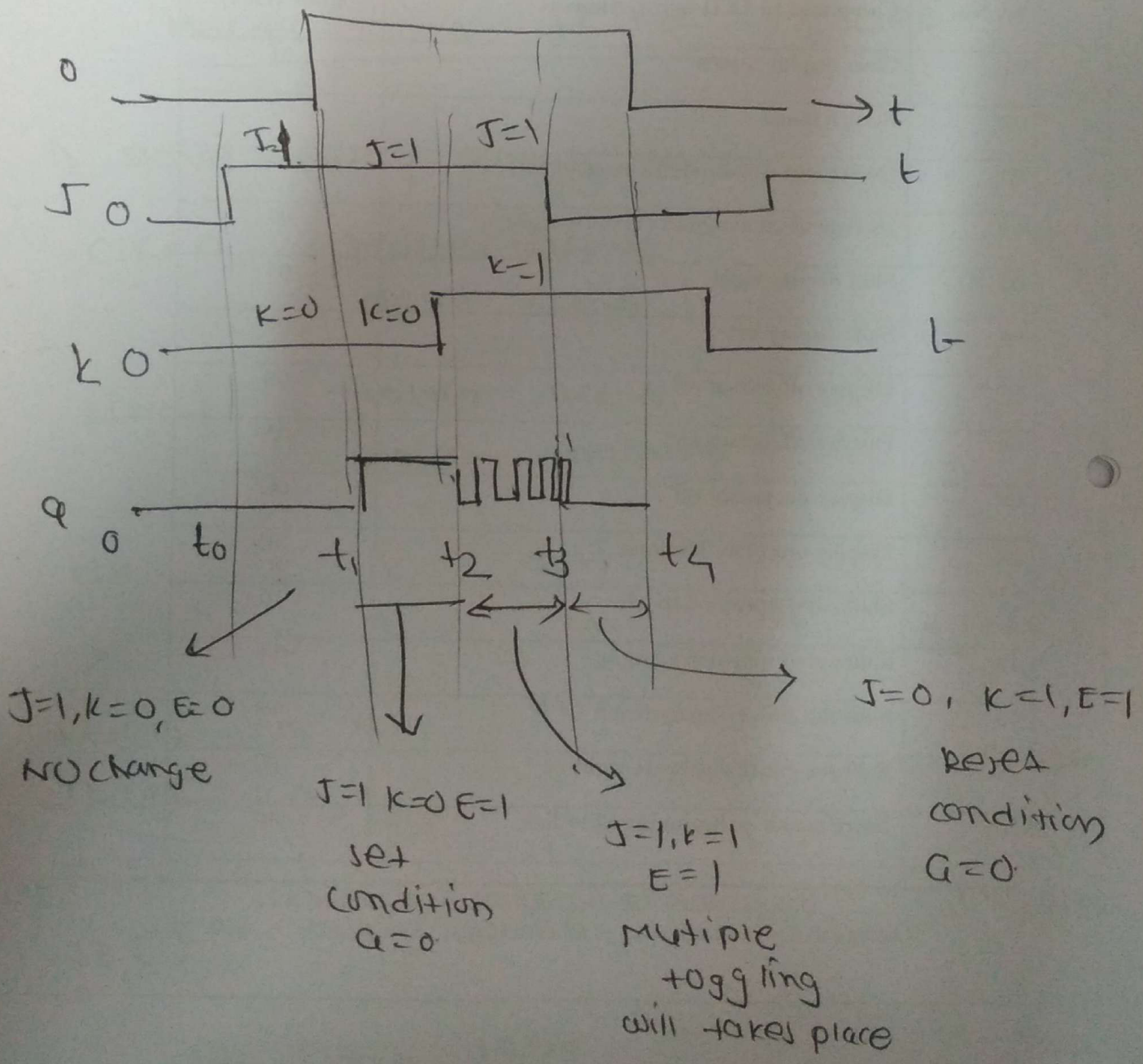
-ve level triggered master

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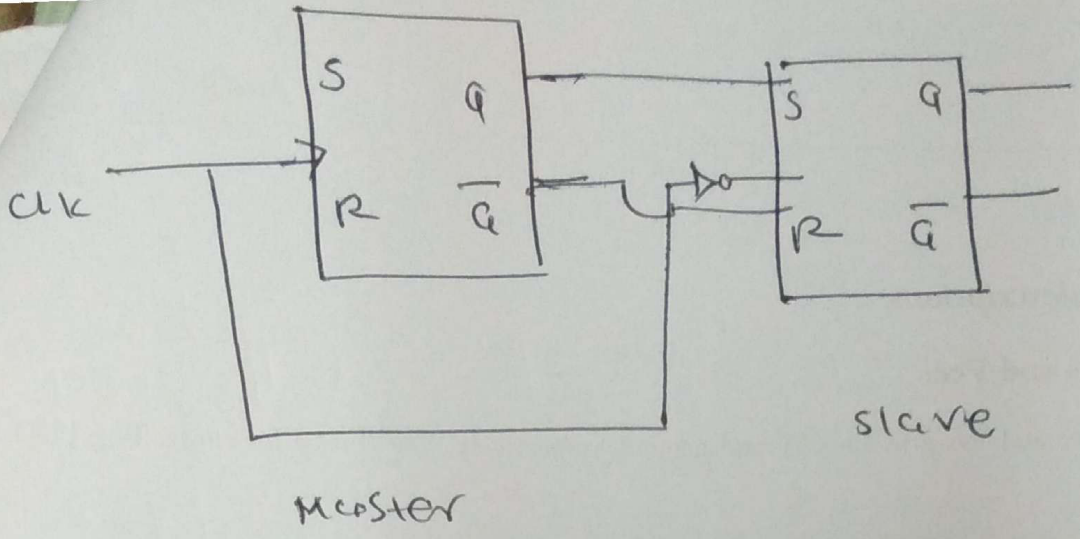
* Race Around condition in JK latch

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	Toggle

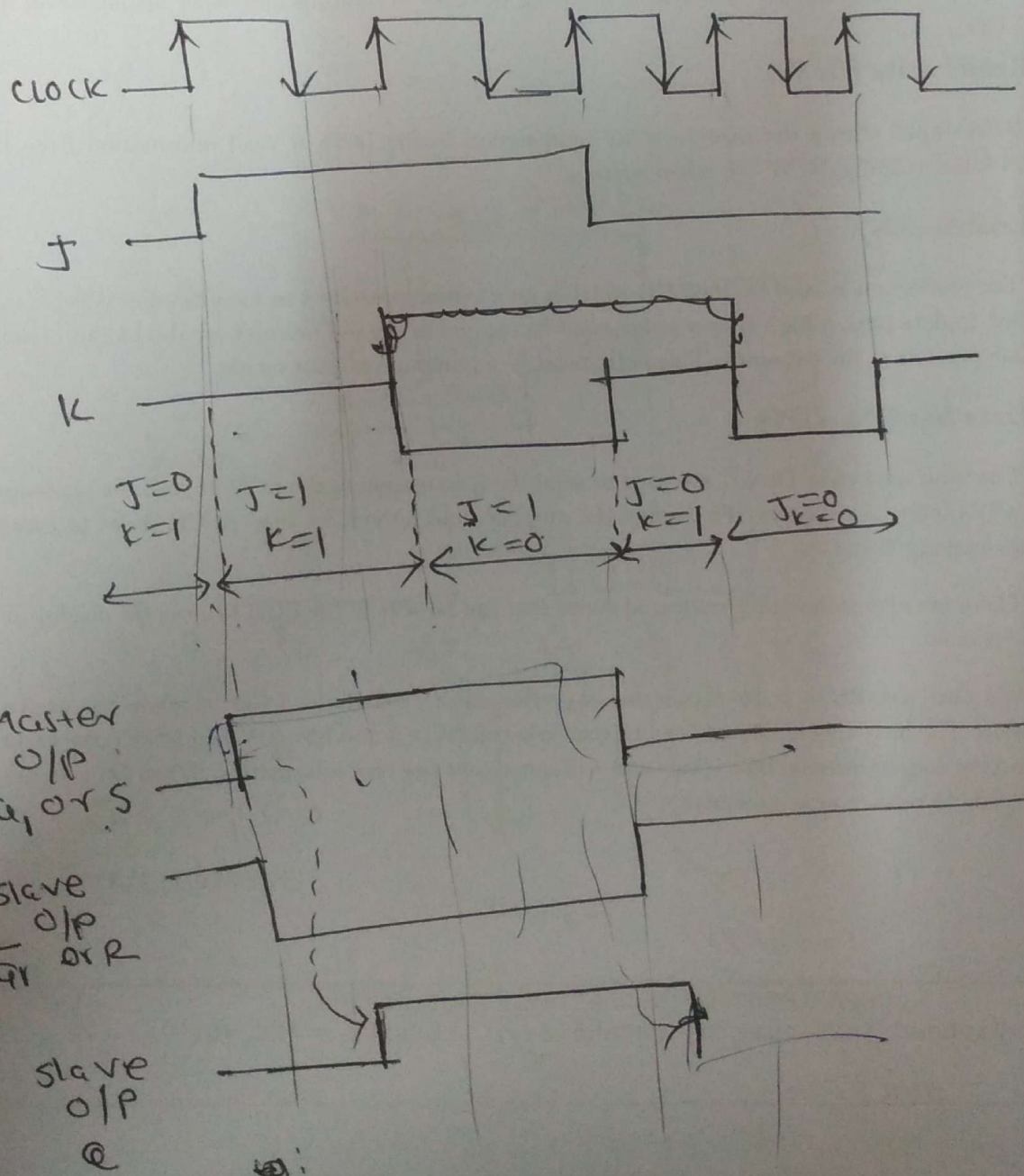


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W/f for Master slave JK F/F

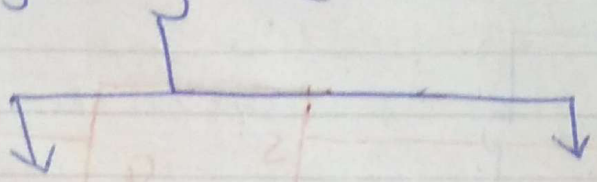


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Edge Triggering :-

triggering method :-

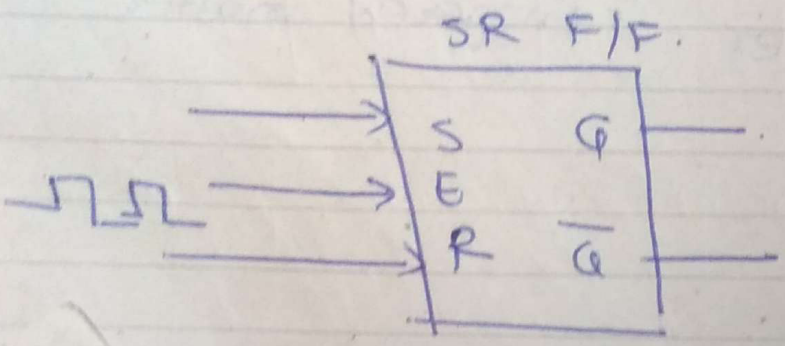


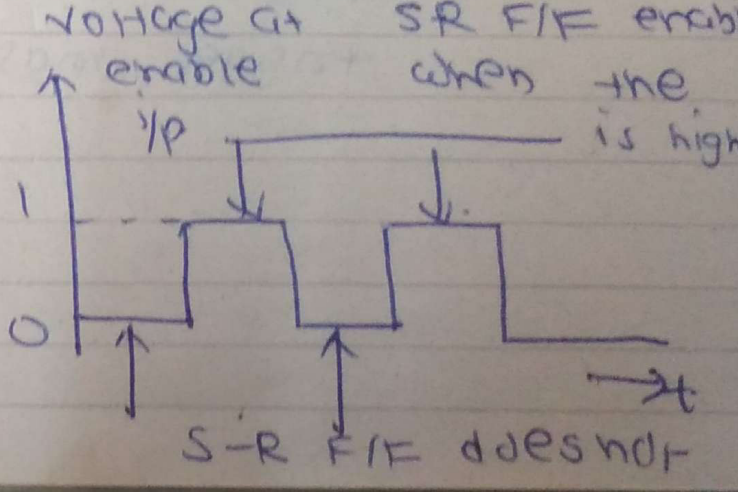
Level triggered Edge triggered

1> Types of level triggering :-

1> +ve level triggered — I/p high (1)

2> -ve level triggered — I/p low (0)





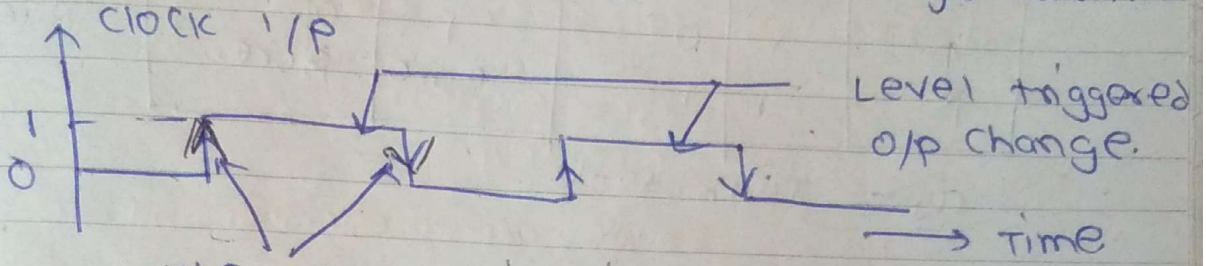
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fig :- level triggering -
Not used practically.

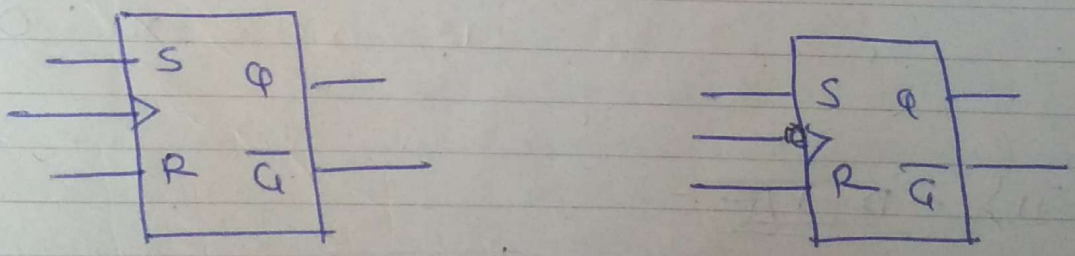
Concept of Edge triggering :-

1) The F/F will change their o/p only corresponding to the +ve & -ve edge of clock. i/p are called as edge triggering F/F.
2) It is called edge triggered or edge sensitive



The edge triggering F/F the o/p will changes due to +ve & -ve edge

Types of edge triggered F/F (Clocked F/F
1) +ve edge triggered
2) -ve edge triggered.



(a) +ve edge triggered (b) -ve edge triggered

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Excitation Table of F/F –:

Excitation Table for S-R F/F –:

CLK	Sn	Rn	Qn+1	$\overline{Qn+1}$
↑	0	0	Qn	$\overline{Qn}$
↑	0	1	0	1
↑	1	0	1	0
↑	1	1	Race	Race

N.S.

0	1
0	0
1	1
Race	

fig –: Truth table

I> S-R F/F –:

present state of Q O/P	Next state Q O/P	Sn I/P	Rn I/P
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

II> D F/F –:

P.S. Q O/P	N.S. Q O/P	I/P Dn.
0	0	0
0	1	1
1	0	0
1	1	1

III> Excitation table for J-K F/F –:

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Q	O/p	I/p
P.S.	N.S.	Tn
0	0	0
0	1	1
1	0	1
1	1	0

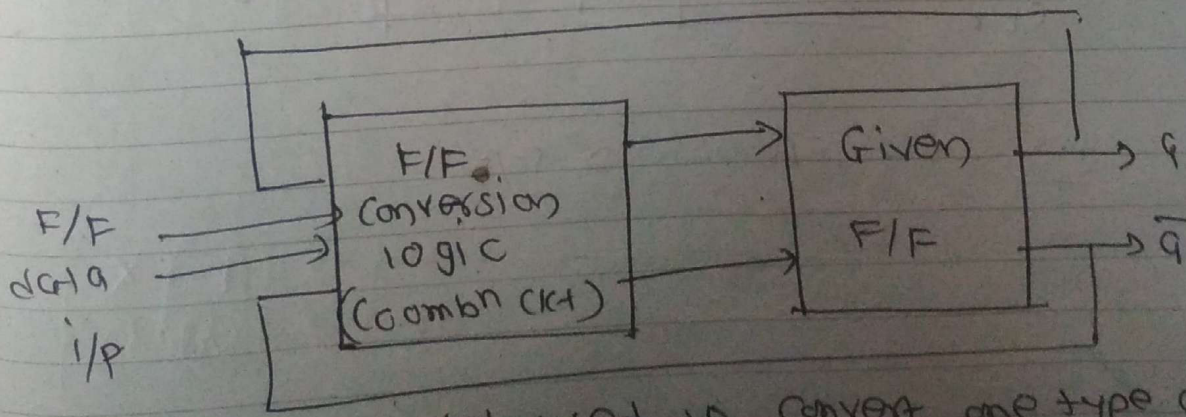
P.S	N.S.	Jn	Kn
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

J	K	Qn	NS.
0	0	0	0
0	1	1	0
1	0	0	1
1	1	1	0

IV) T-F/F.

Q O/p.		I/p
P.S.	N.S.	Tn
0	0	0
0	1	1
1	0	1
1	1	0

* conversion of one type F/F to the other
1) using i) excitation table
2) K-map simplification.
2) Required F/F is combination of given F/F & combinational logic ckt. using gates.
3) I/p to F/F conversion logic are the F/F data i/p & o/p of given F/F.
4) It is designed by using excitation table of both F/F



General model used to convert one type of F/F

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① Conversion from S-R F/F to D F/F.

I/p		N.S.	O/p	
D.	P.S Q_n	Q_{n+1}	S	R
0	0	0	0	x
1	0	1	1	0
0	1	0	0	1
1	1	1	x	0

Entries from excitation table of D F/F

Entries from excitation table of S-R F/F

K-map

For S O/p

D \ Q_n	0	1
0	0	0
1	1	x

$S = D$

For R O/p

D \ Q_n	0	1
0	x	1
1	0	0

$R = \overline{D}$

Logic diagram

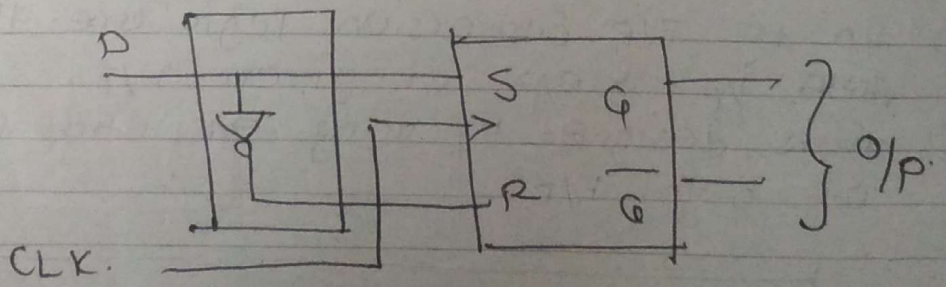


fig :- SR F/F to D F/F

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② conversion from JK F/F to T F/F

T	P.S. Q_n	N.S. Q_{n+1}	O/p J	O/p K
0	0	0	0	X
1	0	1	1	X
1	1	0	X	1
0	1	1	X	0

K-map

for J

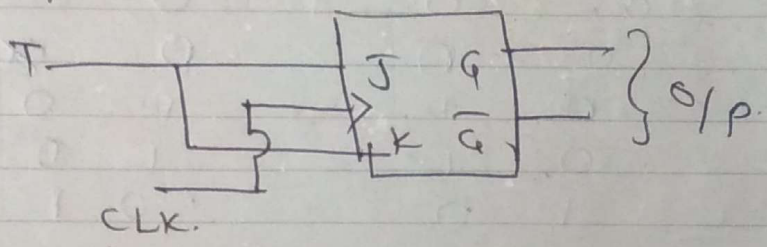
T \ Q_n	0	1
0	0	X
1	1	X

$J = T$.

For K

T \ Q_n	0	1
0	X	0
1	X	1

$K = T$.



③ SR F/F to T F/F.

T	P.S. Q_n	N.S. Q_{n+1}	O/p S	O/p R
0	0	0	0	X
1	0	1	1	0
1	1	0	0	1
0	1	1	X	0

K-map

for S

T \ Q_n	0	1
0	0	X
1	1	0

$S = T \cdot \overline{Q_n}$

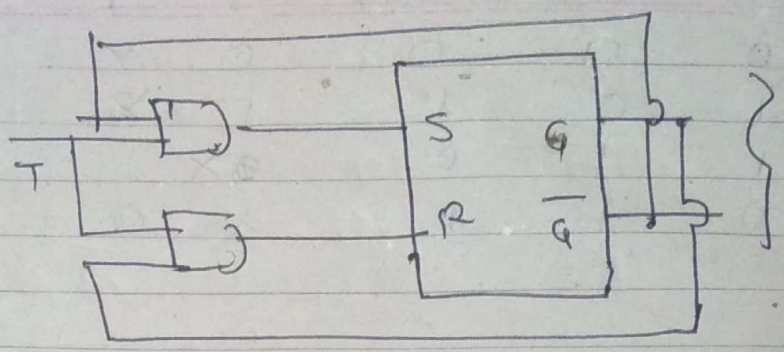
For R

T \ Q_n	0	1
0	X	0
1	0	1

$R = T \cdot Q_n$

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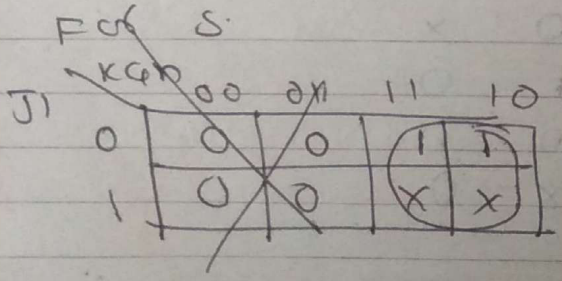
conversion from SR F/F to T F/F.

④ S-R F/F to J-K F/F :-

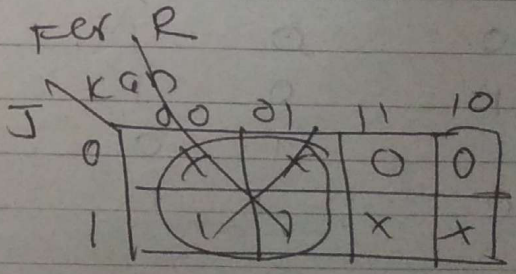
I/P		P.S.	N.S	O/P	
J	K	Q_n	Q_{n+1}	S	R
0	0	0	0	0	x
0	1	0	0	0	0
1	0	0	1	1	0
1	1	0	1	1	0
0	1	1	0	0	1
1	1	1	0	0	1
0	0	1	1	x	0
0	1	1	1	x	0

K-map

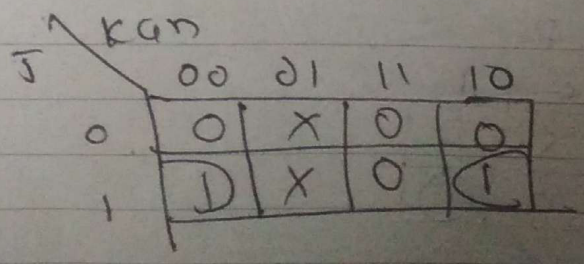
For S



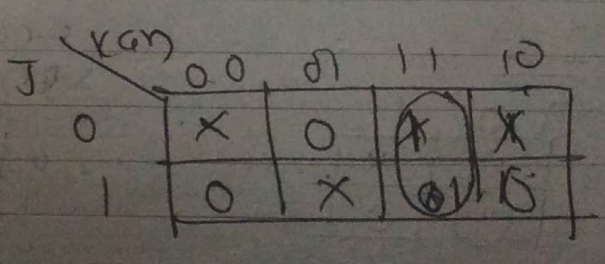
For R



$S = K$



$R = \bar{K}$



$S = \bar{J} \cdot \bar{Q_n}$

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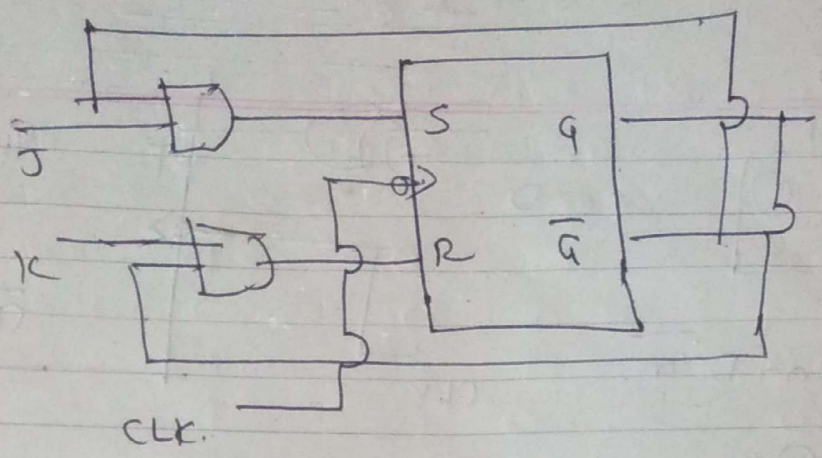
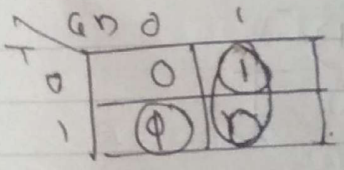


fig - S-R F/F to JK F/F. conversion.

⑤ Conversion from D F/F to T F/F.

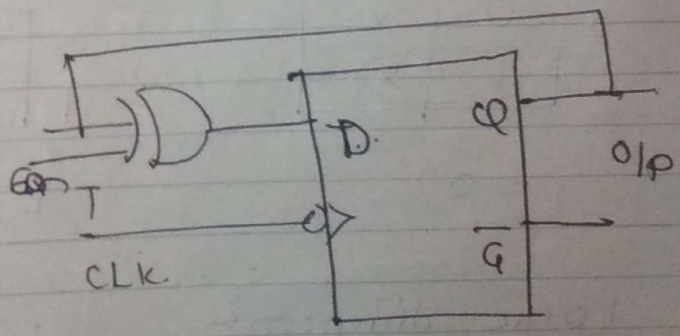
T	P.S. Q_n	N.S. Q_{n+1}	D F/F
0	0	0	0
1	0	1	1
1	1	0	0
0	1	1	1

K-map :- D F



$D = Q_n \cdot \bar{T} + T \cdot \bar{Q_n}$

Logic diagram



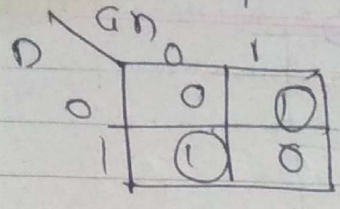
⑥ T F/F to D F/F conversion :-

D	P.S. Q_n	N.S Q_{n+1}	O/P. T
0	0	0	0
1	0	1	1
0	1	0	1
1	1	1	0

https://t.me/SPPU_TE_BE_COMP

Scanned by CamScanner

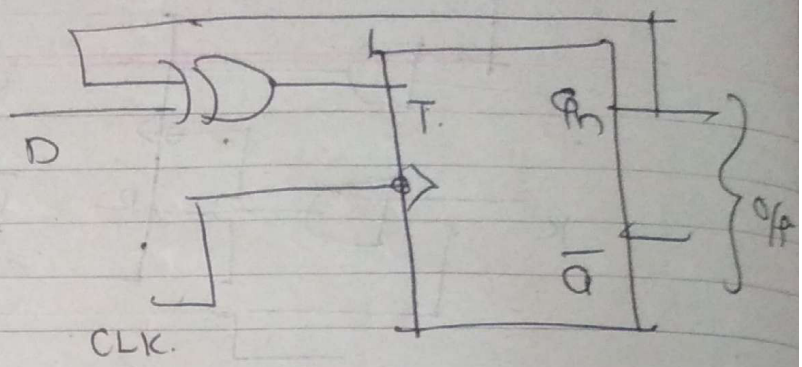
For T.



$T = \bar{D} \cdot Q_n + D \cdot \overline{Q_n}$

$T = D \oplus Q_n$

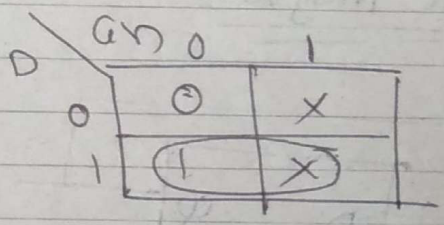
Logic diagram :-



(7) JK to D F/F conversion :-

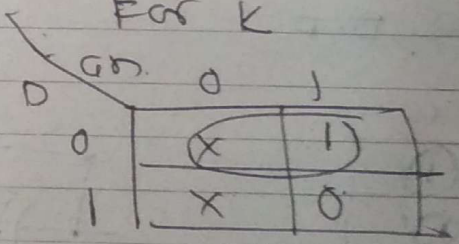
D	P.S. Qn	N.S. Qn+1	O/P J	K
0	0	0	0	x
1	0	1	1	x
0	1	0	x	1
1	1	1	x	0

K-map for J.



$J = D$

For K



$K = \bar{D}$

Logic diagram :-

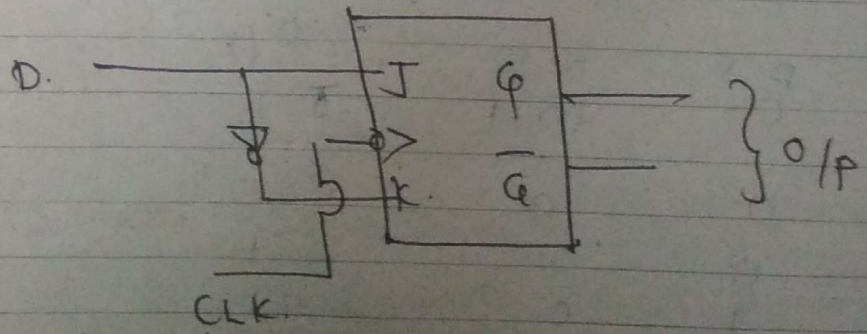
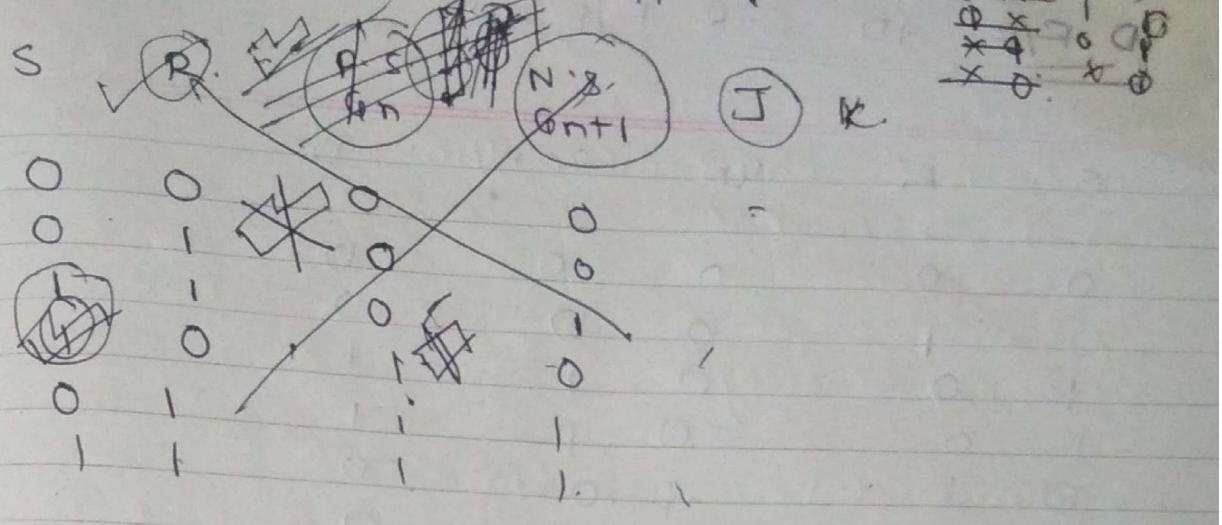


fig :- Logic diagram from conversion of JK F/F to D F/F

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S	R	P.S. Q_n	N.S. Q_{n+1}	J	K
0	0	0	0	0	X
0	1	0	0	0	X
1	0	0	1	1	X
1	0	0	1	1	X
0	1	1	0	X	1
0	1	1	0	X	1
0	0	1	1	X	0
1	0	1	1	X	0

K-map

For J

S \ RQn	00	01	11	10
0	0	0	0	0
1	1	X	X	X

$J = S$

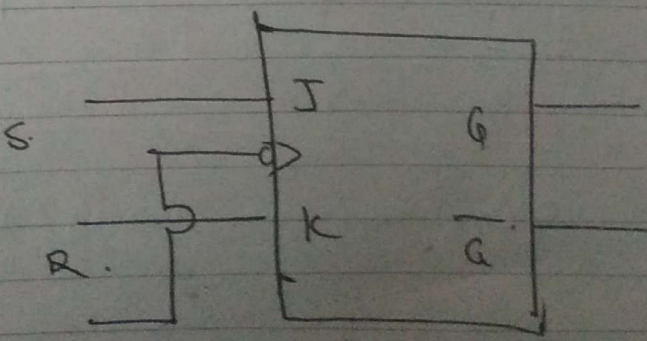
For K

S \ RQn	00	01	11	10
0	X	0	1	X
1	X	0	X	X

$K = R$

Logic diagram :-

(If condition will not occur use 'x' in k map)



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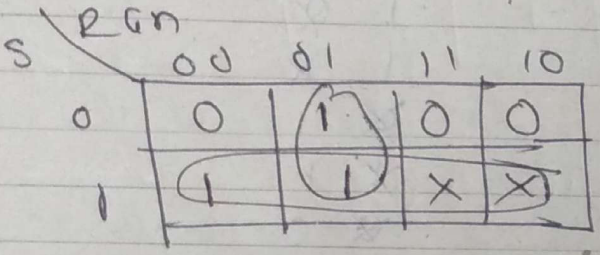
Scanned by CamScanner

D F/F to S-R F/F

S	R	Q_n	Q_{n+1}	D
0	0	0	0	0
0	1	0	0	0
1	0	0	1	1
1	0	0	1	1
0	1	1	0	0
0	1	1	0	0
0	0	1	1	1
1	0	1	1	1

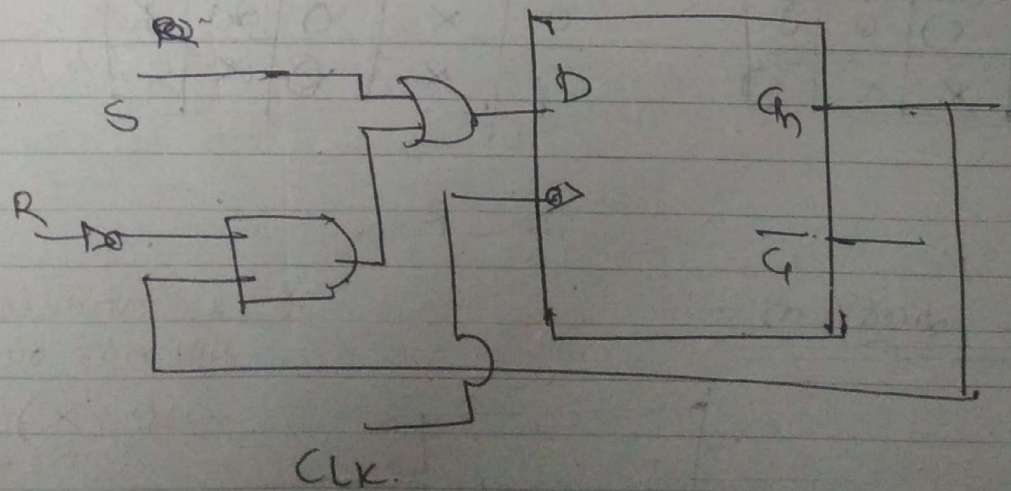
K-map

For D



$D = S + \bar{R}Q_n$

Logic diagram



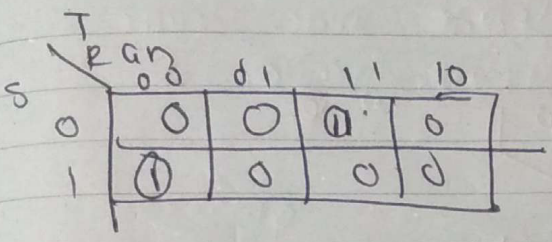
https://t.me/SPPU_TE_BE_COMP

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T F/F to SR

S	R.	Q_n	Q_{n+1}	T.
0	x	0	0	0
1	0	0	1	1
0	1	1	0	1
x	0	1	1	0

K-map for T



$T = \bar{S}R.Q_n + S.\bar{R}\bar{Q_n}$

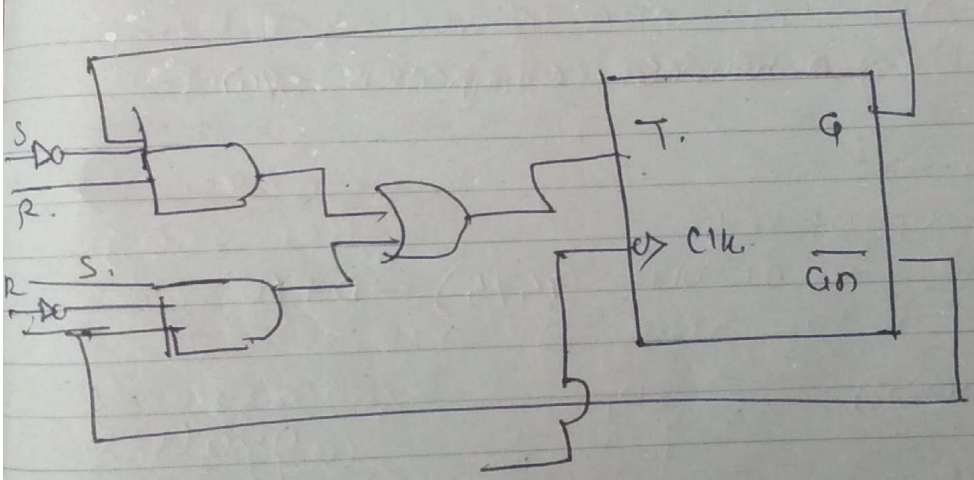


fig - T F/F to S-R

* If no. of I/p are two then we 8 cells of k-map.

* If no. of I/p are only one then we 4 cell of k-map

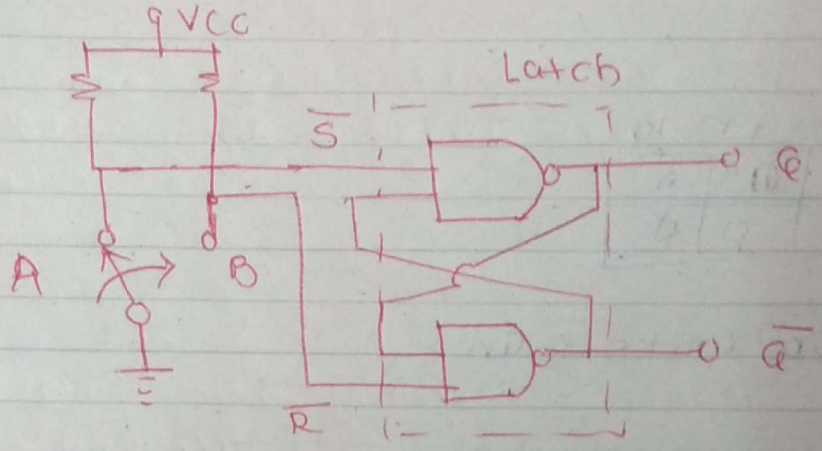
https://t.me/SPPU_TE_BE_COMP

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Application of F/F

1) Bounce elimination s/w
2) register
3) counter.
4) Latch
5) RAM / memory.

(1) Bounce elimination s/w :-



(a) A bounce elimination s/w

(2) Register :-

1) It is group of F/F
2) It stores no. of bit (7474) - D F/F

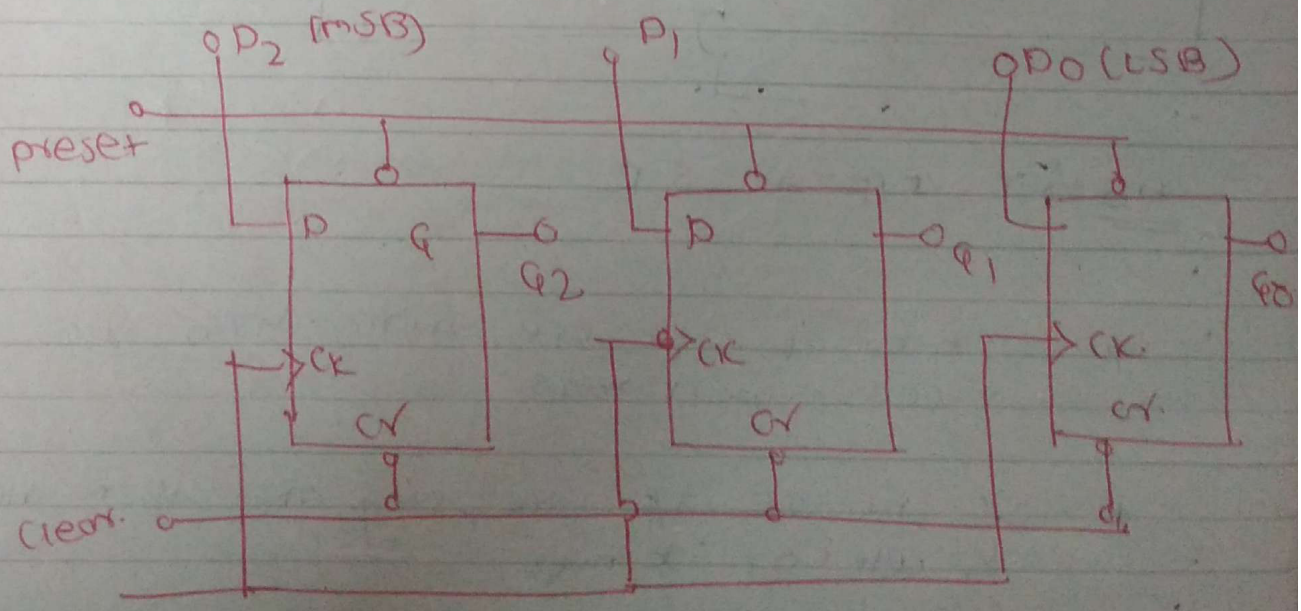
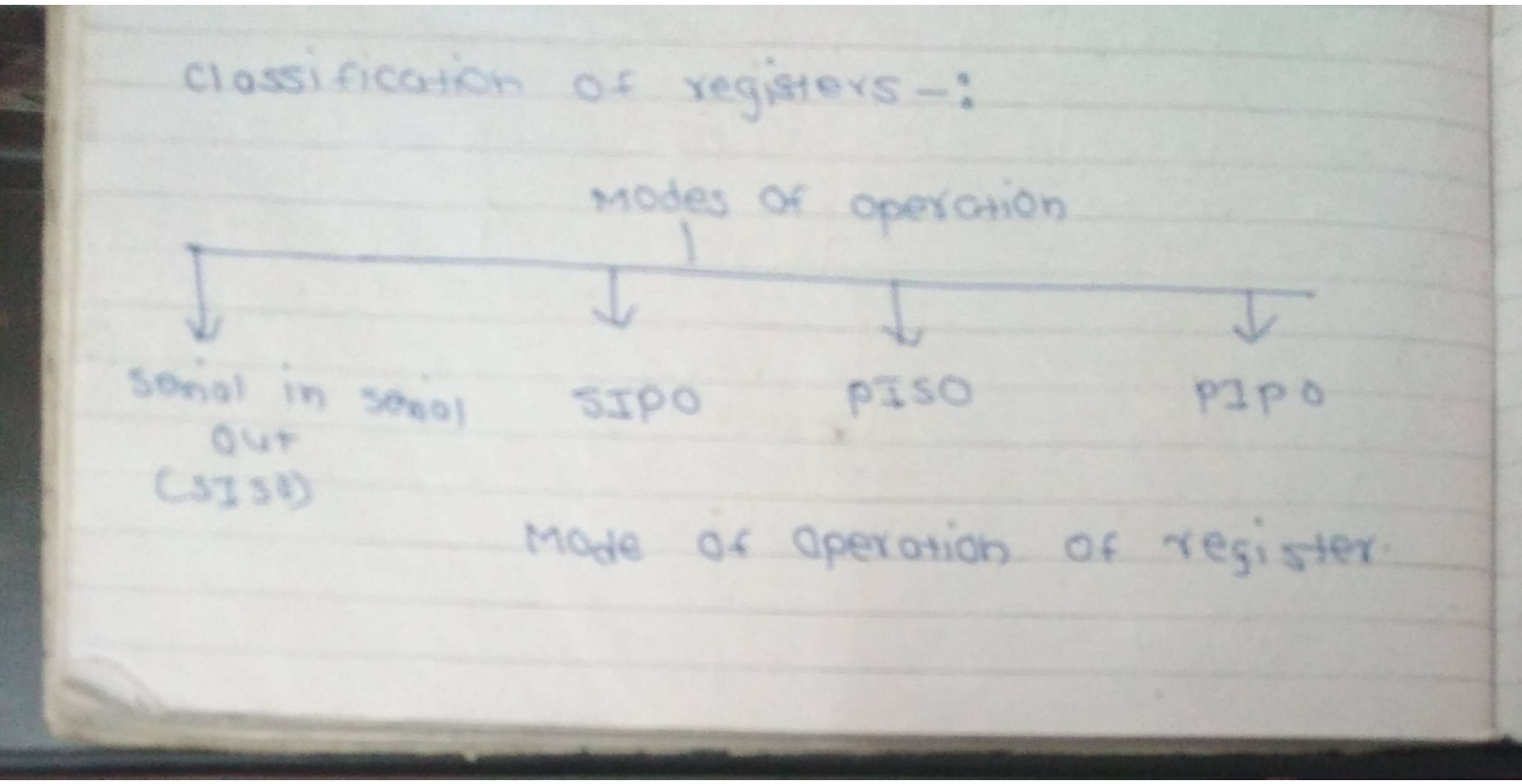


fig :- A 3-bit register using F/F.

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Classification of registers -:

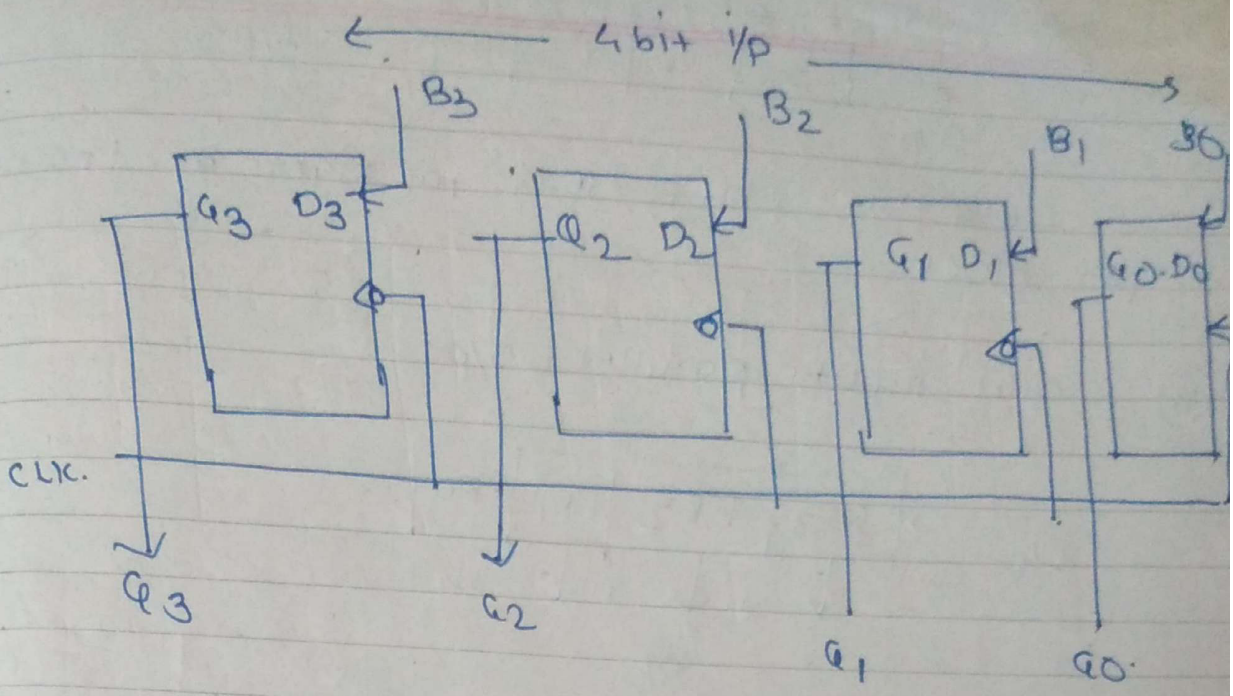


Mode of operation of register

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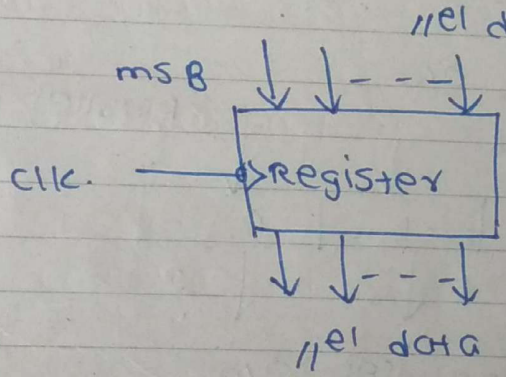
Buffer registers :-



A four bit buffer register using D F/F.

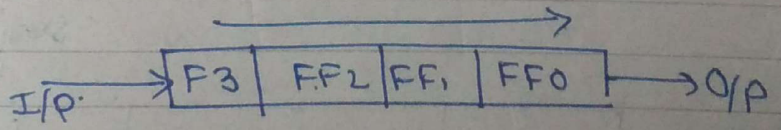
$Q_3 Q_2 Q_1 Q_0 = B_3 B_2 B_1 B_0 = 1010$

Schmatic of buffer register :-



Shift registers :-

1) serial input serial output (serial shift right)

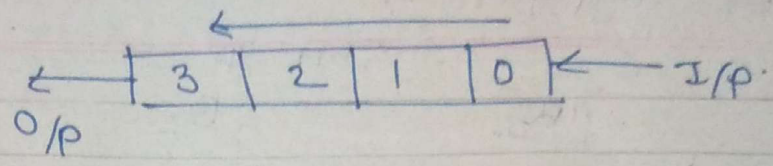


Data bit shift from left to right by one position.

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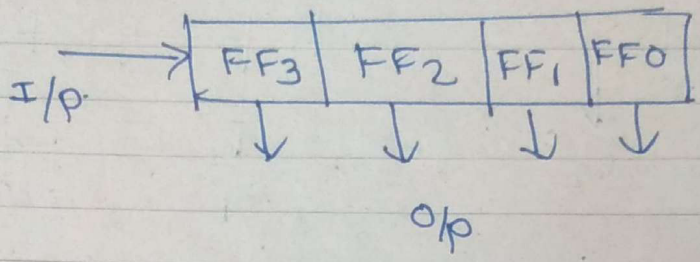
Scanned by CamScanner

② serial i/p serial o/p (shift left)

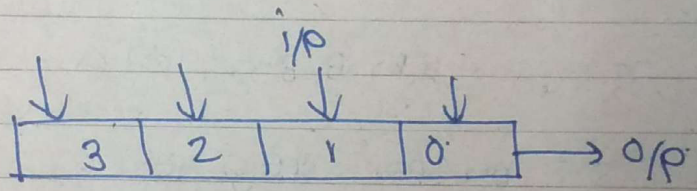


Data bit shift left to right by 1 position per clock.

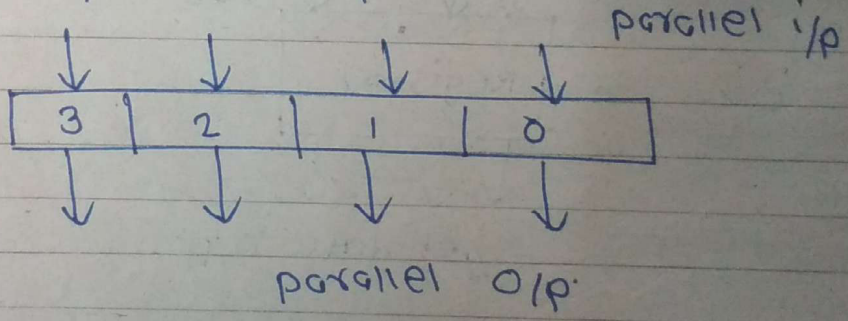
③ serial input parallel o/p



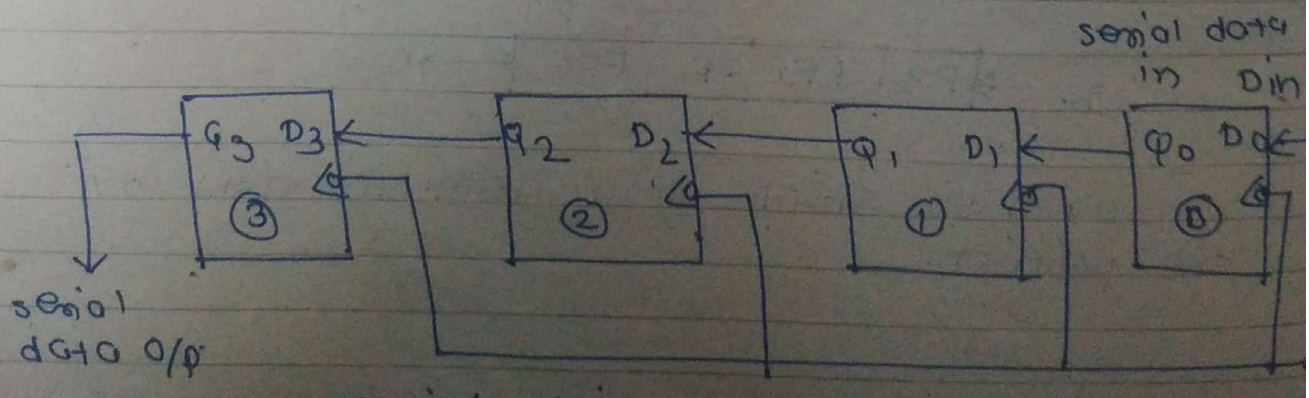
④ parallel input serial o/p



⑤ ||el i/p parallel o/p



I) serial input serial output (shift left mode)



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$\Rightarrow$ $Q_3Q_2Q_1Q_0 = 0000$ $\longrightarrow$ Reset

$Q_3Q_2Q_1Q_0 = 0001$

$Q_3Q_2Q_1Q_0 = 0011$ (Apply MSB bit first)

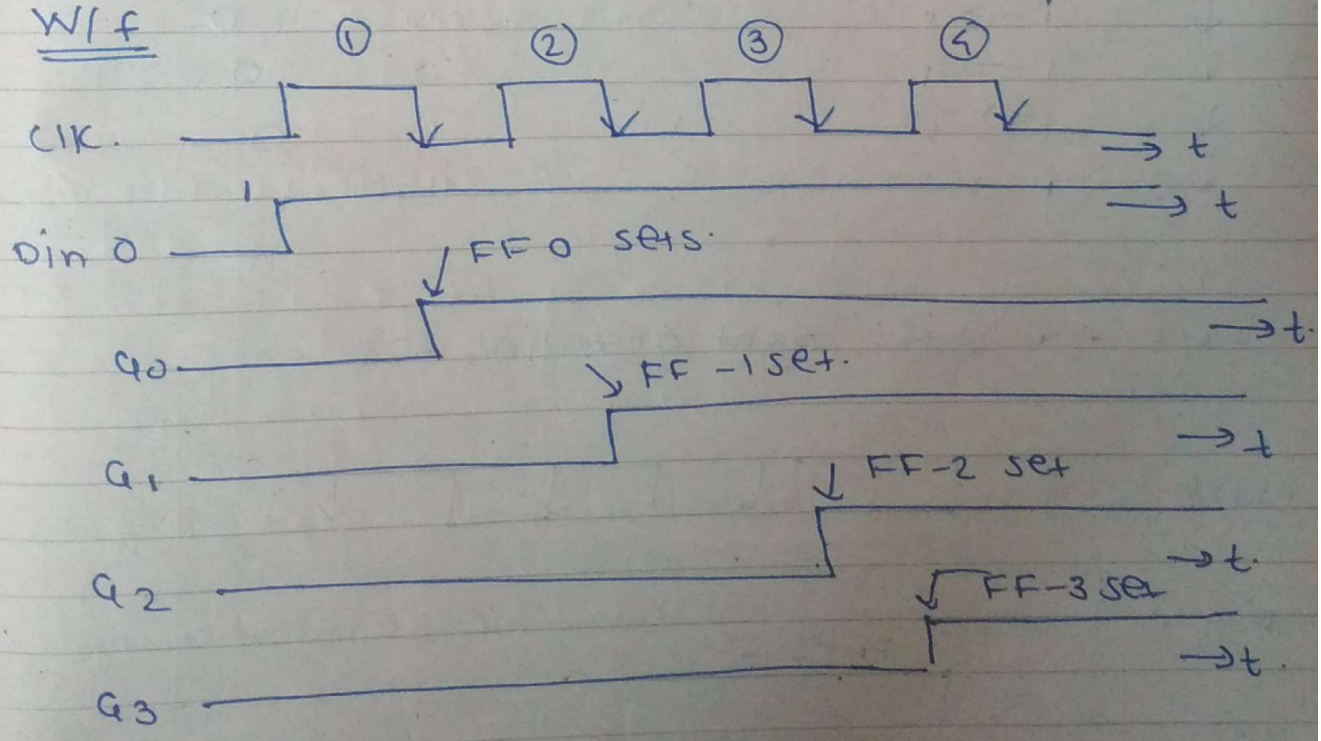
$= 0111$

$Q_3Q_2Q_1Q_0 = 1111$

Clk. Initially	Q_3	$Q_2 = D_3$	$Q_1 = D_2$	$Q_0 = D_1$, $Din = D_0$
↓	0	0	0	0
↓	0	0	0	1 ← 1
↓	0	0	1	1 ← 1
↓	0	1	1	1 ← 1
↓	1	1	1	1 ← 1

← Direction of travel.

W/f

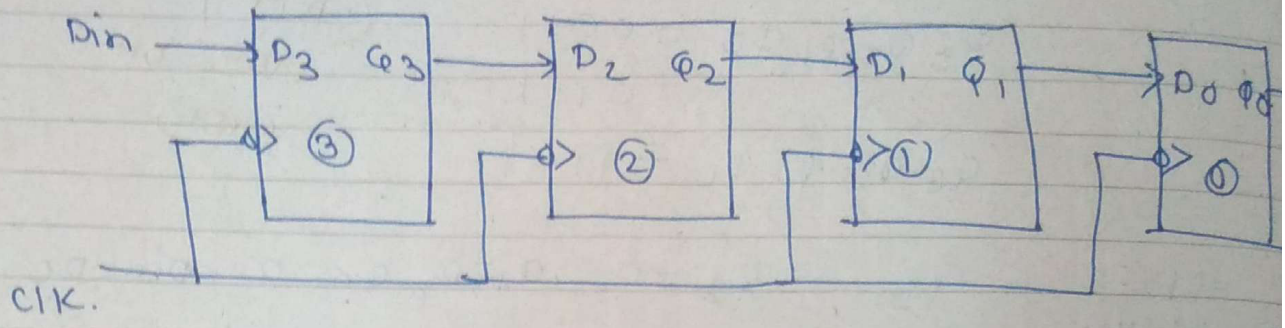


II) serial in serial out (shift right mode :-

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SISO (shift right mode)

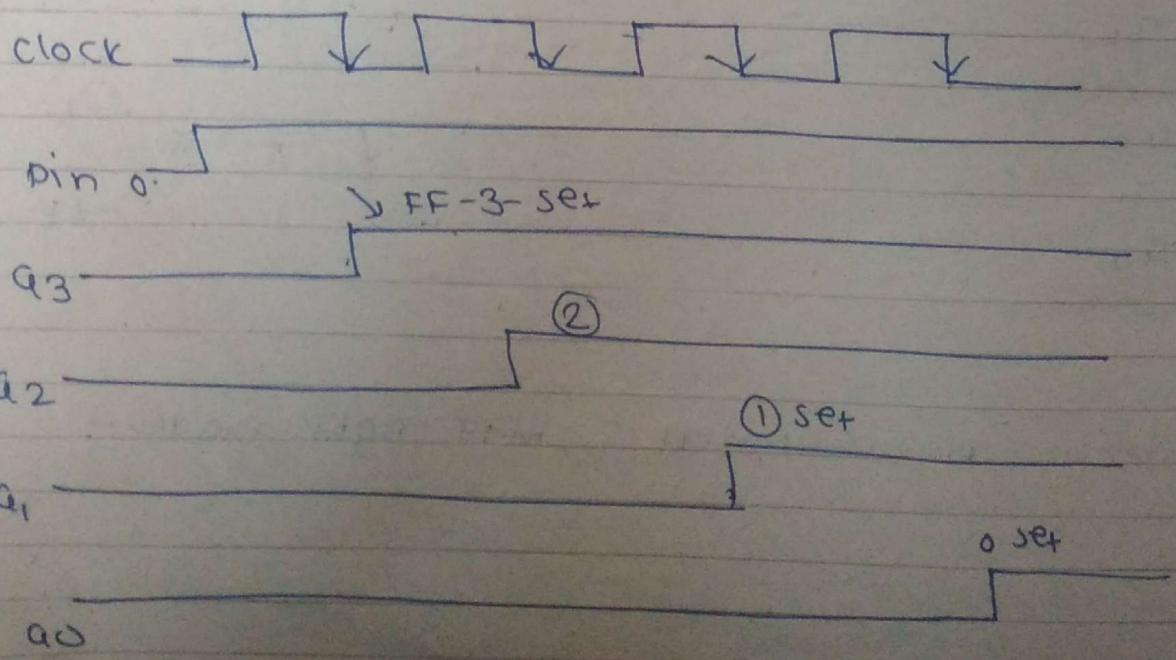


serial shift right register.

$Q_3 Q_2 Q_1 Q_0 = 0000$ (Apply LSB bit first)

CLK.	Din = Q3	Q3 = D2	Q2 = D1	Q1 = D0	Q0
	⓪D	0	0	0	
↓	1 → 1	0	0	0	
↓	1 → 1	1	0	0	
↓	1 → 1	1	1	0	
↓	1 → 1	1	1	1	

w/f for shift right operation :-



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II) serial In parallel out :-

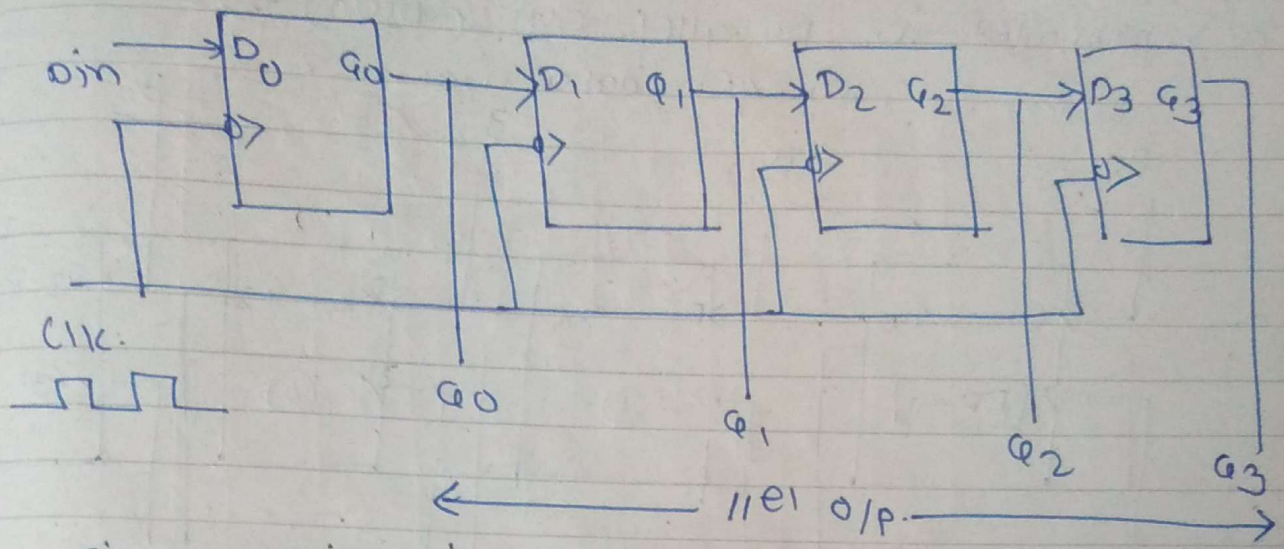
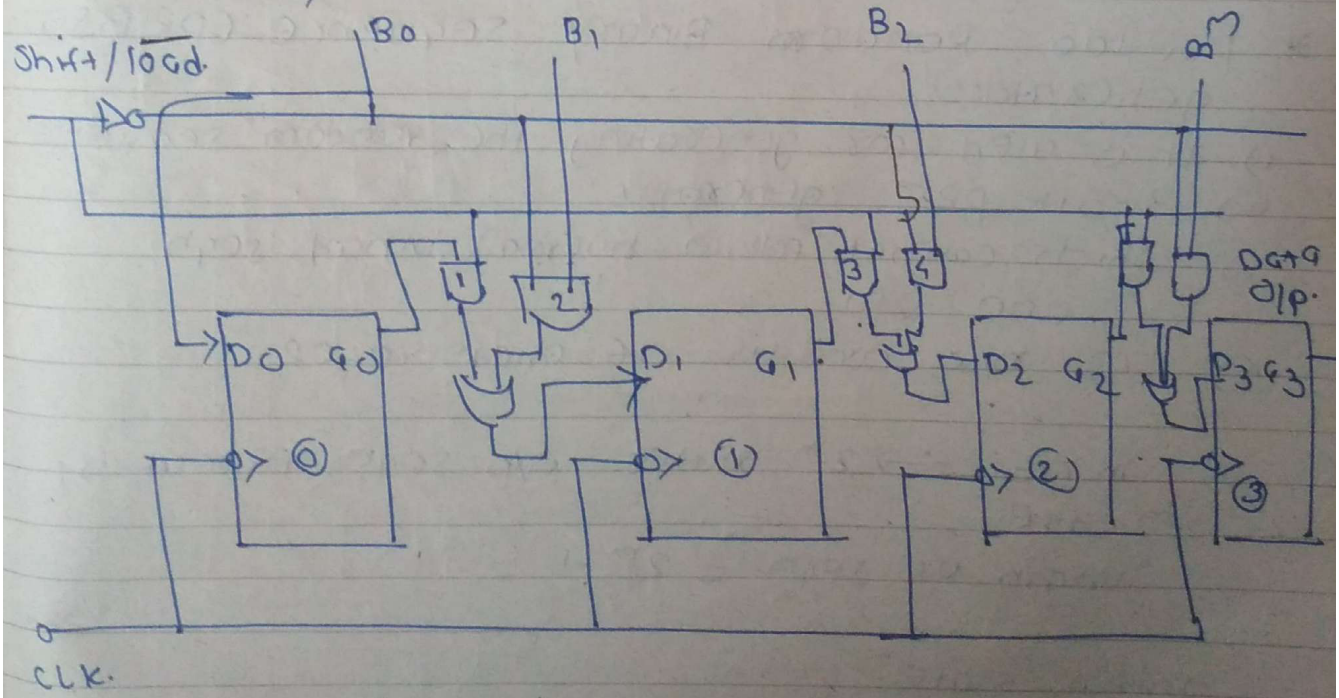


fig :- serial in parallel output mode.

III) parallel in serial out mode (PISO) :-
operates on two mode
1) shift mode
2) load mode



Parallel in serial out shift register

Load mode :-
shift/load = 0. Parallel loading.

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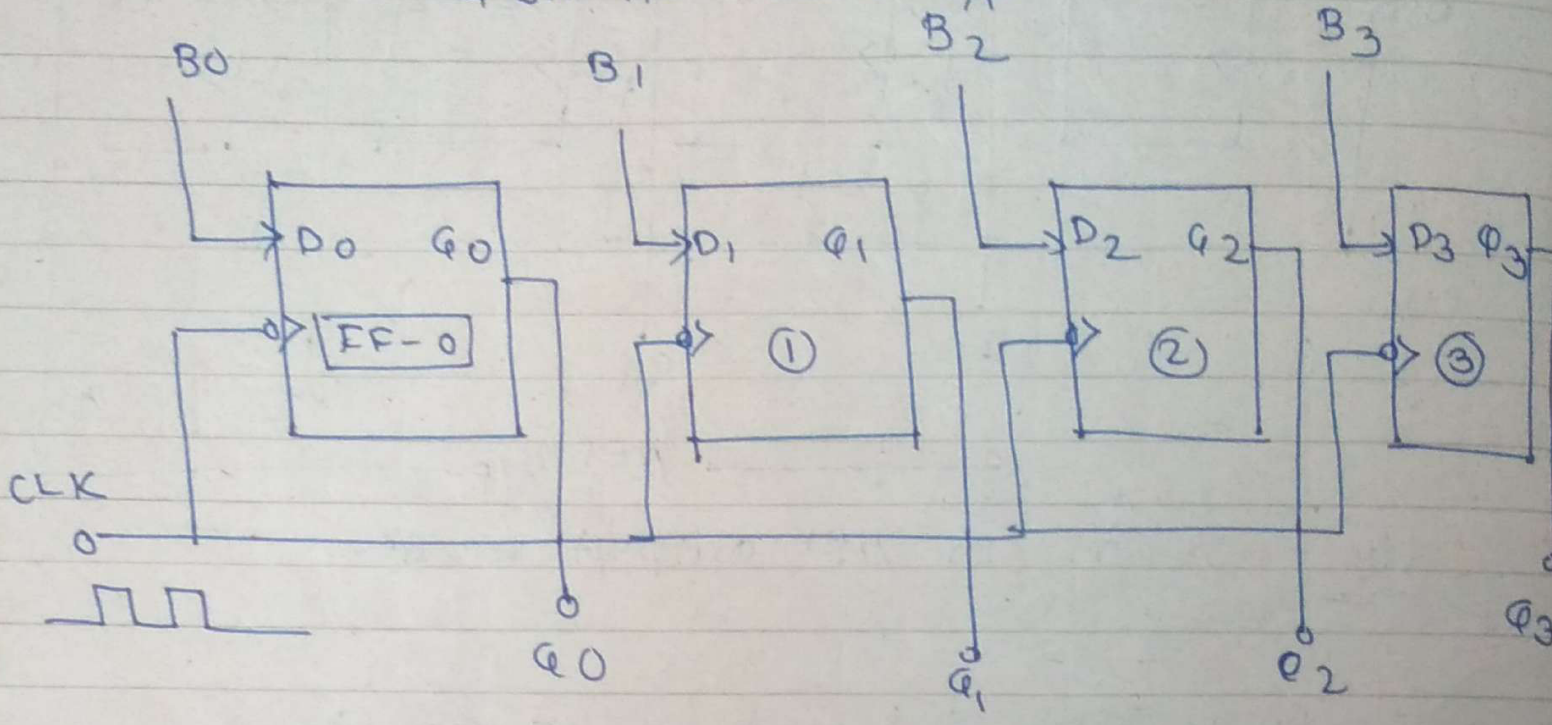
Shift mode :-

Shift / $\overline{\text{load}}$ = high (1)

left to right operation.

IV) parallel in parallel out (PIPO) :-

4 bit //el data i/p.



//el in //el out shift register.

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*Bidirectional shift register :-

1) using D
2) using J-K.

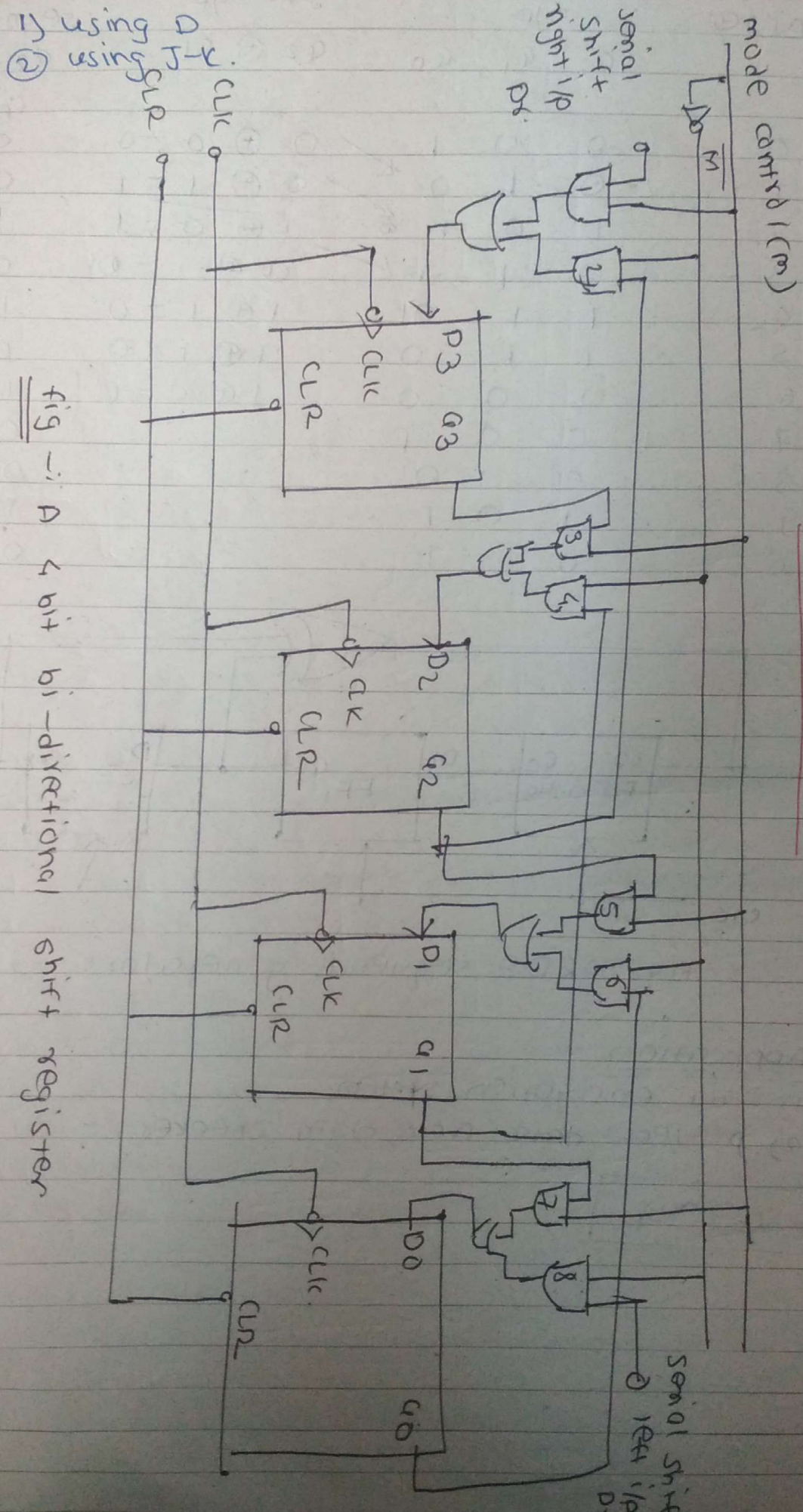


fig :- D 4 bit bi-directional shift register

3-bit using J-K (duo)

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operation -: multiply 2

① Mode = 1 shift right operation.

AND gate. 1, 3, 5, 7 enable & remaining AND gate, 2-4, 6, 8 disabled

FF3 to FF0

② Mode = 0 shift left operation.

2, 4, 6, 8 = enabled

1, 3, 5, 7 = disabled

FF0 to FF3

M should be changed when clk = 0.

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shift register (right except for one change

* Ring counter -

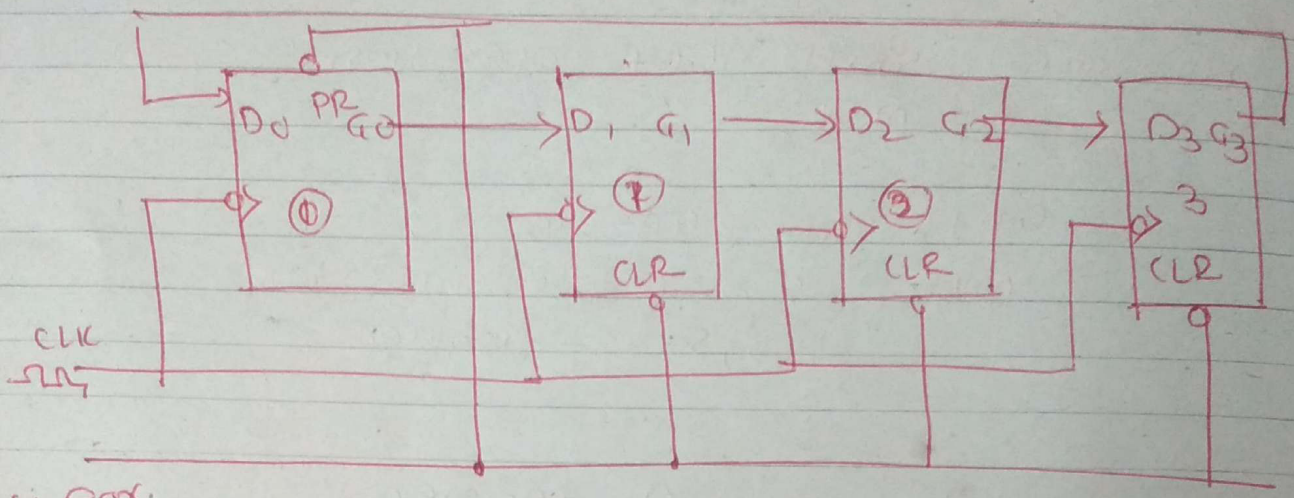


fig :- A four bit ring counter

Q3 Q2 Q1 Q0 = 0001
= 0010
= 0100
= 1000
= 0001

operation

CLR	CLK	Q0	Q1	Q2	Q3	
⊓	x	1	0	0	0	← F/F 0 preset other clear
1	↓	0	1	0	0	
1	↓	0	0	1	0	
1	↓	0	0	0	1	
1	↓	1	0	0	0	

using J-K

$J_0 = Q_3$

$K = \overline{Q_3}$

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Johnson's counter (Twisted / switched Tail Ring counter) -:

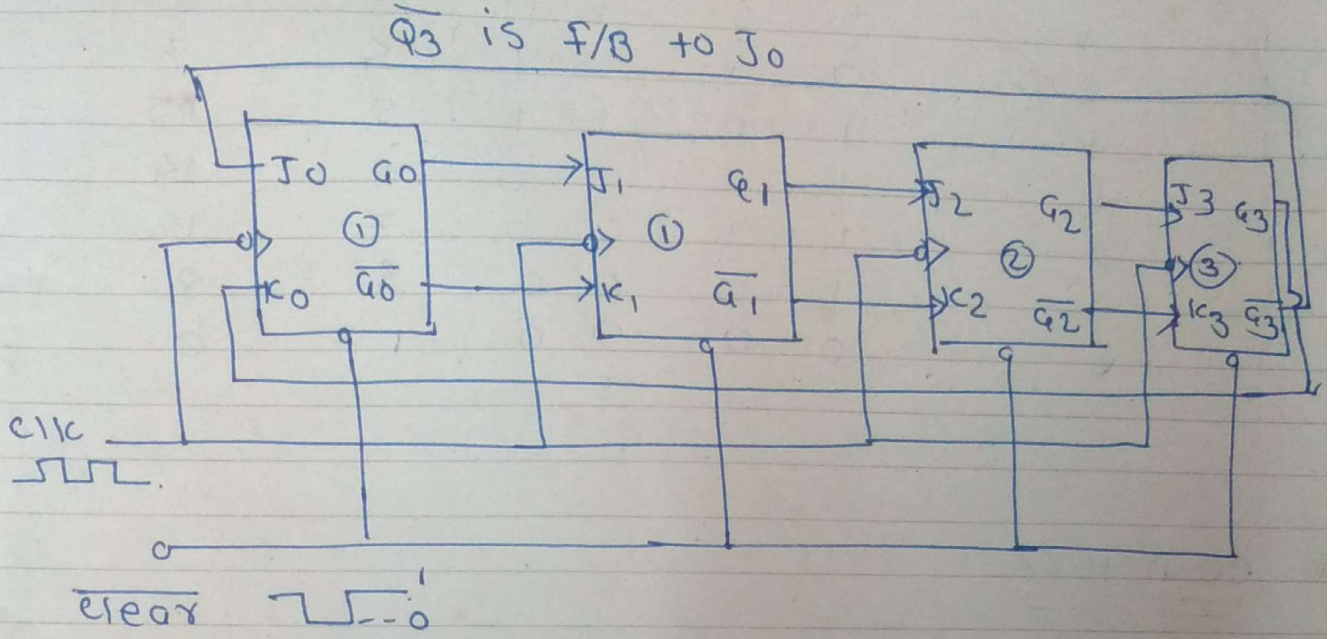


fig -: Twisted ring counter or Johnson counter.

Operation -:

$Q_3 Q_2 Q_1 Q_0 = 0000$

$Q_3 Q_2 Q_1 Q_0 = 0001$

$Q_3 Q_2 Q_1 Q_0 = 0011$

$Q_3 Q_2 Q_1 Q_0 = 0111$

$Q_3 Q_2 Q_1 Q_0 = 1111$

$Q_3 Q_2 Q_1 Q_0 = 1110$

$Q_3 Q_2 Q_1 Q_0 = 1100$

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Clear	CLK.	Q_3	Q_2	Q_1	Q_0	State no.	Decimal eqn
⊓	Initially	0	0	0	0	1	0
1	↓	0	0	0	1	2	1
1	↓	0	0	1	1	3	3
1	↓	0	1	1	1	4	7
1	↓	1	1	1	1	5	15
1	↓	1	1	1	0	6	14
1	↓	1	1	0	0	7	12
1	↓	1	0	0	0	8	8
1	↓	0	0	0	0	1	0

counters :-

1) Asynchronous (Ripple) up counter
2) Asynchronous down counter
3) UP/DOWN counter.

Intd :-

1) Digital ckt. used for counting pulses is known as counter.
2) Application of F/F.

Types of counter
1) Asynchronous or ripple counter
2) synchronous counter

Asynchronous or ripple counter :-
For these o/p of counter clock s/g is applied to one F/F & then o/p of preceding F/F is connected to clock to next F/F.

synchronous counter :-
In the synchronous counter all the F/F receive the clock pulse simultaneously. Ring counter & Johnsons counter example of synchronous coun

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"Asynchronous" means ripple.

Asynchronous / Ripple up counters :-

no. of F/F = no. of bit

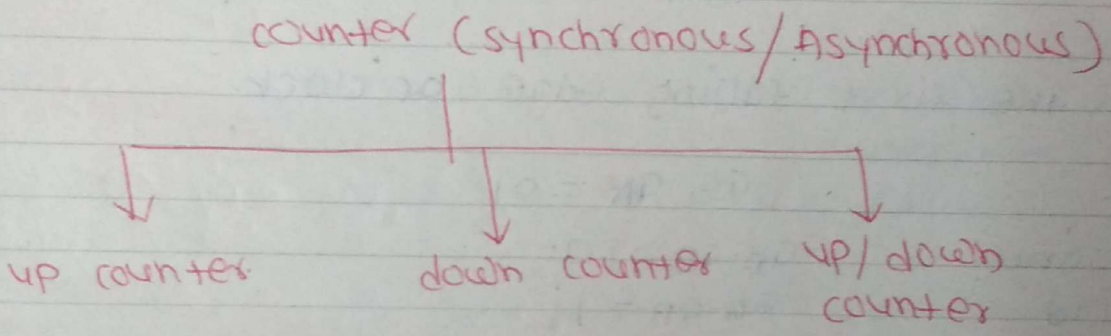
1) 2-bit Asynchronous / ripple counter
2) 3-bit Asynchronous / ripple counter
3) 4-bit Asynchronous counter

1) 2-bit Asynchronous counter :-

using J=K=1 & T=1

Classification of counter :-

Based on counting progress



up counter :-

The counter that counts from small to big. The o/p goes on increasing as they receive the clock pulse.

e.g :- o/p of up counter 0-1-2-3.

Down counter :-

The counter that counts from small to big count. The o/p goes on decreasing as they receive clock pulses.

e.g. :- o/p of down counter is 7-6-5-4-3

up/down counter :-

It is combination of up counter & down counter.

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Asynchronous / Ripple counter :-

1) 2-bit

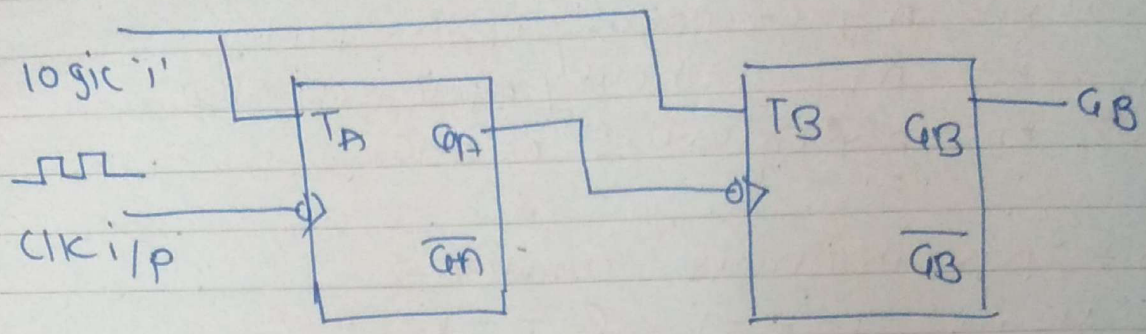


fig :- 2-bit up counter.

operation :-

Initiall all F/F reset

$Q_B Q_A = 00$

on the first falling edge of clock.

$Q_B Q_A = 01$

second $Q_B \cdot Q_B = 10$.

Third $Q_A \cdot Q_B = 11$

The both the stage don't occur simultaneously is called Asynchronous counter

Truth table :-

Clock	counter o/p. Q_B (MSB)	Q_A (LSB)	State Number	Decimal eqn of counter o/p
Initially	0	0	—	0
↓	0	1	1	1
↓	1	0	2	2
↓	1	1	3	3
↓	0	0	4	0

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3-bit Asynchronous up counter :-

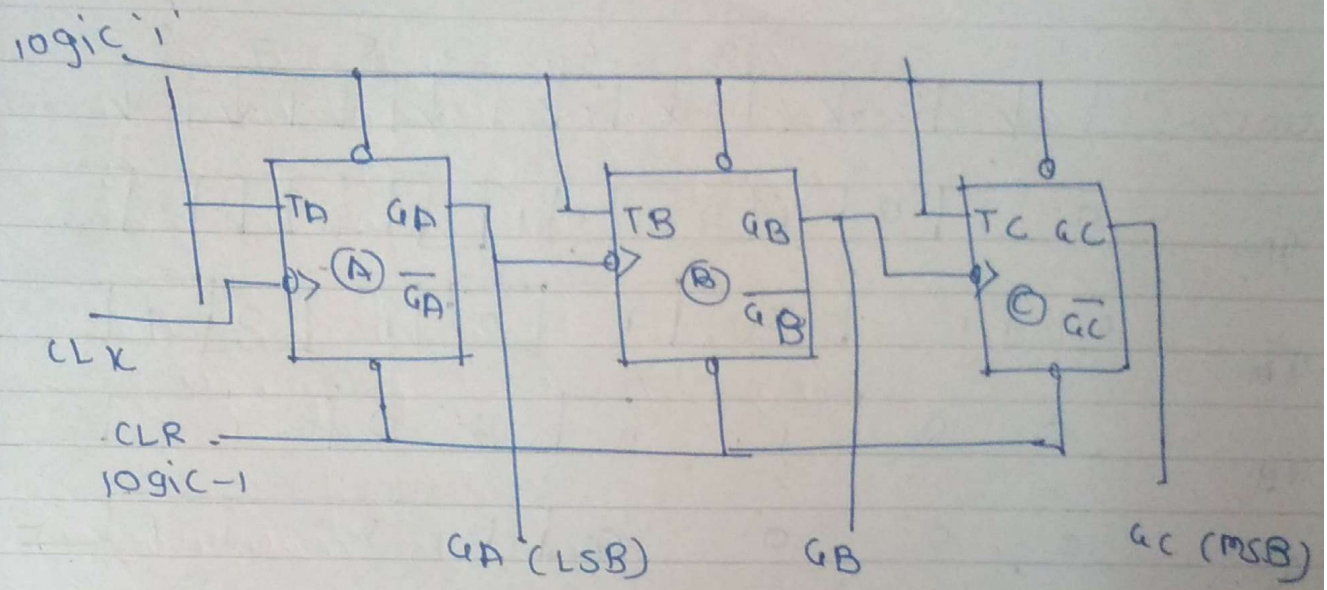


fig :- 3-bit ripple up counter.

Clock	F/F o/p			State	Decimal equivalent
	QC	QB	QA		
Initially	0	0	0	1	0
↓	0	0	1	2	1
↓	0	1	0	3	2
↓	0	1	1	4	3
↓	1	0	0	5	4
↓	1	0	1	6	5
↓	1	1	0	7	6
↓	1	1	1	8	7
↓	0	0	0	1	0

Number of state :-

Number of states $= 2^n = 2^3 = 8$

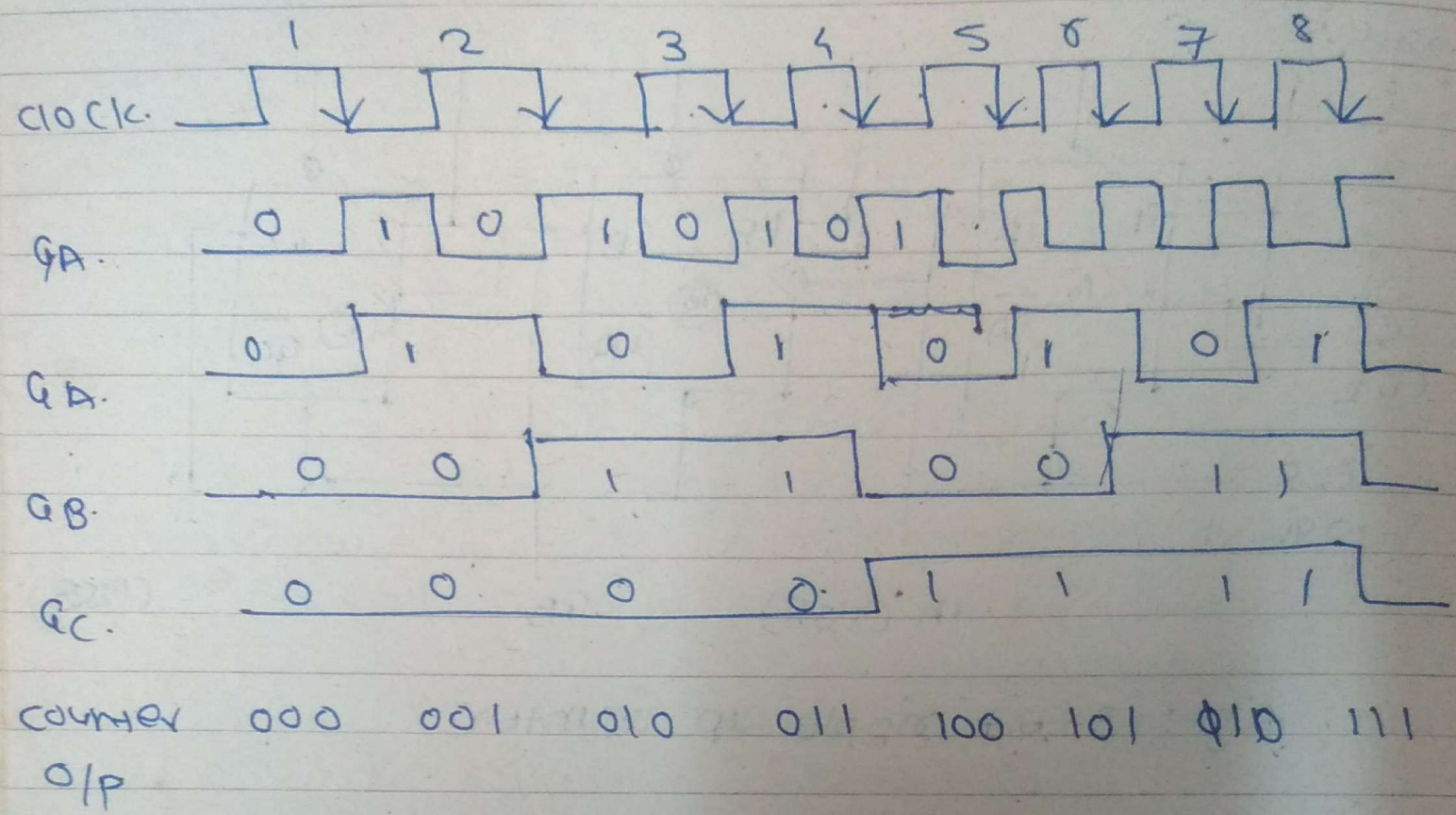
Maximum count :-

maximum count $= 2^{n-1} = 2^3 - 1 = 8 - 1 = 7$

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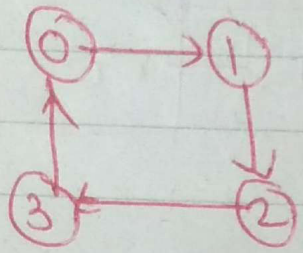
Timing diagram-: (3-bit) ripple up counter.



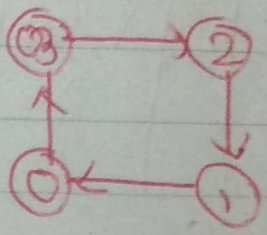
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State diagram of a counter :-
Circle represents state number.



(a) 2-bit up counter.



(b) 2-bit down counter.

State diagram.

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Asynchronous down counter:-

I) 3-bit down counter:-

	Q_C	Q_B	Q_A (F/F o/p)
7	1	1	1
6	1	1	0
5	1	0	1
4	1	0	0
3	0	1	1
2	0	1	0
1	0	0	1
0	0	0	0

Recycle

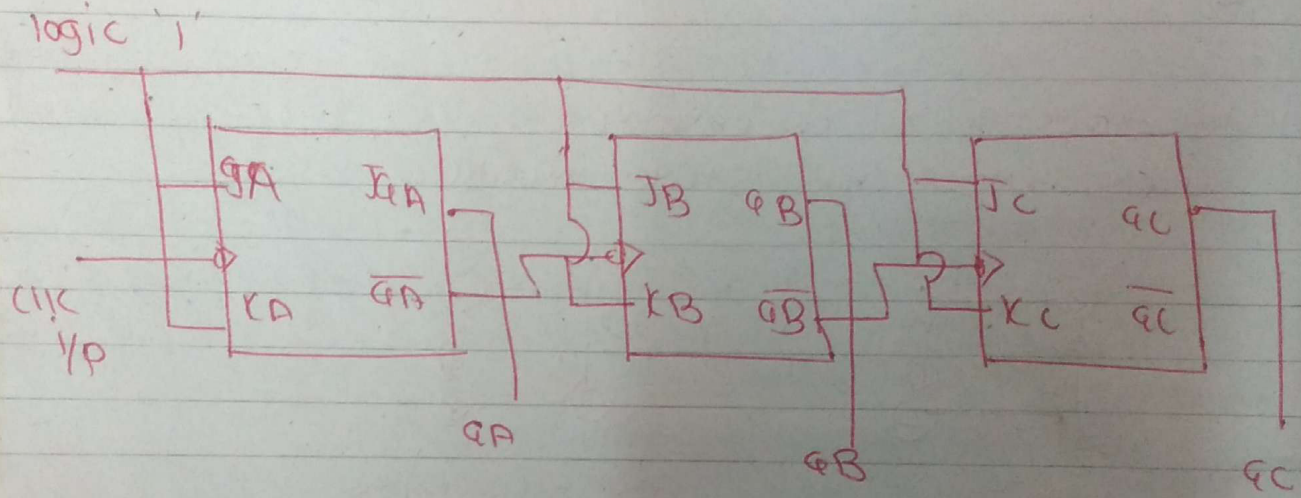


fig :- 3-bit asynchronous down counter

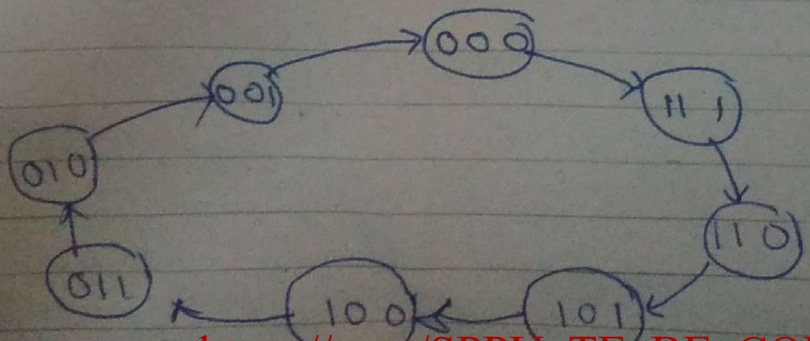
operation:-

$Q_C Q_B Q_A = 000$

$Q_C Q_B Q_A = 111$

$Q_C Q_B Q_A = 110$

state diagram



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Modulus of counter –:

Mod no. = 2^n = No. of states

counter type	Modulus
2 bit up/down	MOD-4
3 bit up/down	MOD-8
4 bit up/down	MOD-16

Modulus of counter represent No. of states.

* Design Decade (BCD) Ripple counter –:

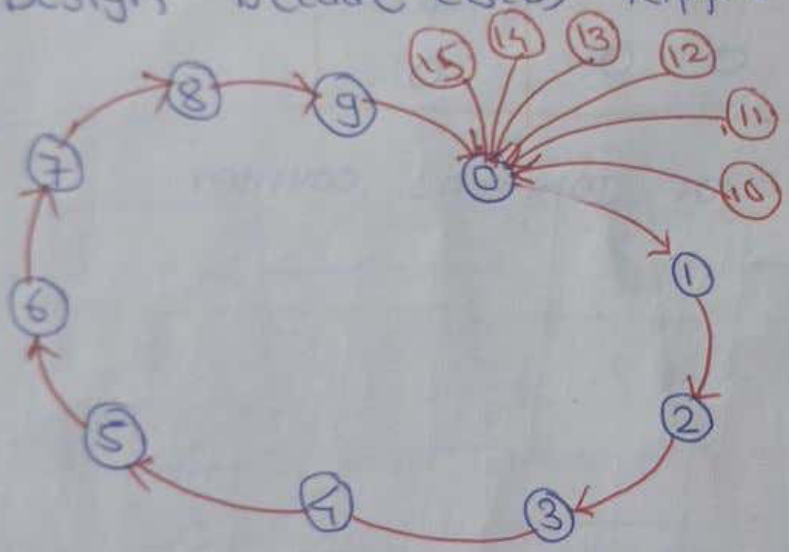


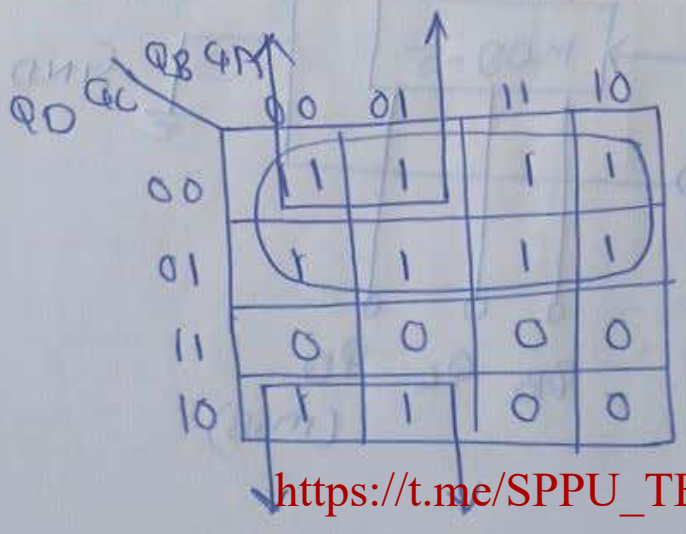
Fig – State diagram of a BCD counter.

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Truth table

CLK	Q_D	Q_C	Q_B	Q_A	O/p of Reset logic	
↓	0	0	0	0	1	Valid states
↓	0	0	0	1	1	
↓	0	0	1	0	1	
↓	0	0	1	1	1	
↓	0	1	0	0	1	
↓	0	1	0	1	1	
↓	0	1	1	0	1	
↓	0	1	1	1	1	
↓	1	0	0	0	1	
↓	1	0	0	1	1	
↓	1	0	1	0	0	Invalid states.
↓	1	0	1	1	0	
↓	1	1	0	0	0	
↓	1	1	0	1	0	
↓	1	1	1	0	0	
↓	1	1	1	1	0	

K-map :-



$$Y = \overline{Q_D} + \overline{Q_C} \cdot \overline{Q_B}$$

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Logic diagram :-

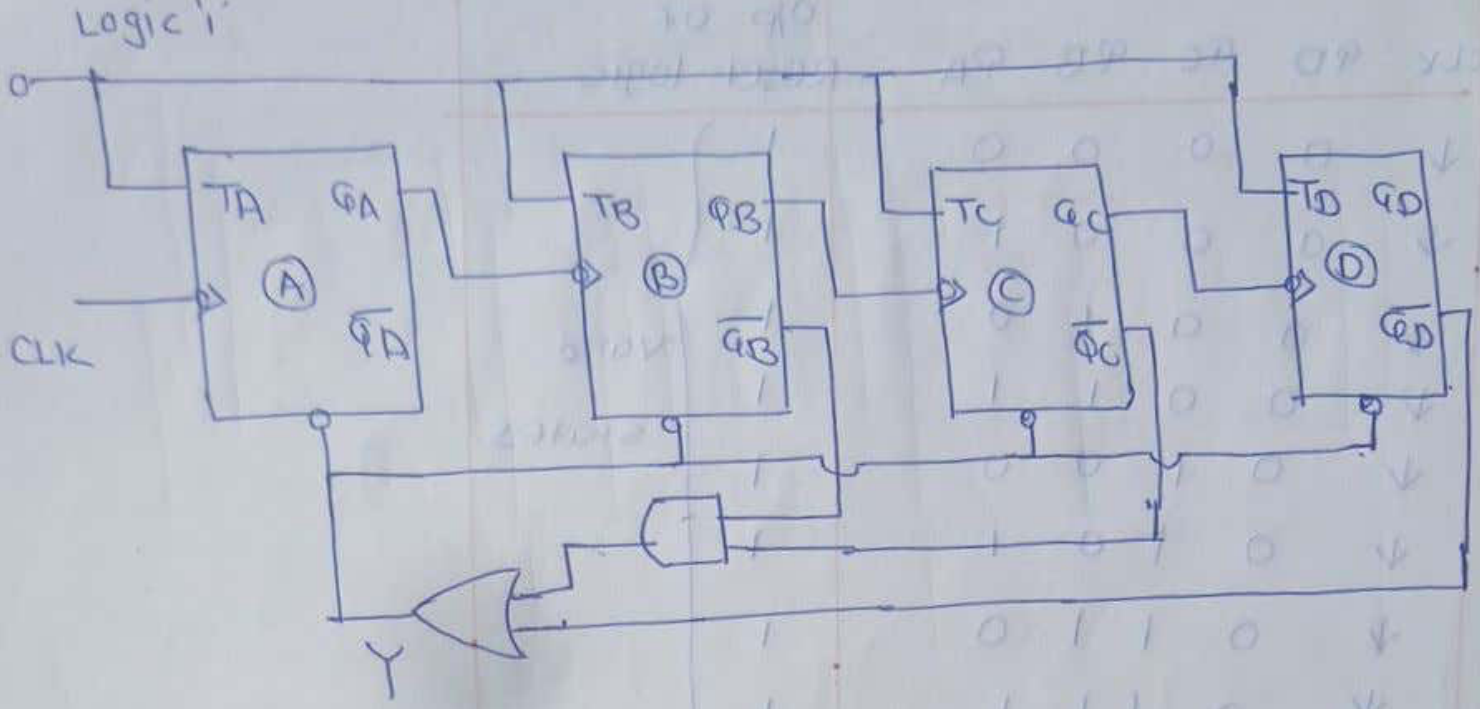


Fig:- Logic diagram of BCD counter (Ripple) (Asynchronous)

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Synchronous Counters :-

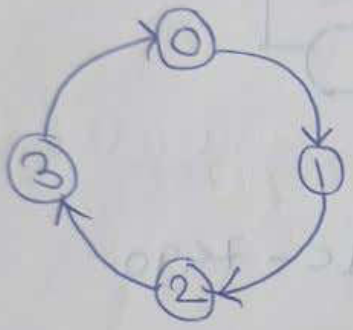
I) 2-bit II) 3-bit III) 4-bit

I) 2-bit synchronous up counter :-

2 bit = MOD-4

Que:- 2-bit synchronous up counter using MS-JK F/F. (4M- Dec 2019)

⇒ 2 bit = MOD-4



(a) State diagram

(b) Excitation table of J-K

P.S. Q_n	N.S. Q_{n+1}	I/P. J	I/P. K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

(c) Circuit Excitation Table :-

P.S. Q_B	P.S. Q_A	N.S. Q_{B+1}	N.S. Q_{A+1}	F/F I/P. J_B	F/F I/P. K_B	F/F I/P. J_A	F/F I/P. K_A
0	0	0	1	0	x	1	x
0	1	1	0	1	x	x	1
1	0	1	1	x	0	1	x
1	1	0	0	x	1	x	1

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(d) K-map :-

J_B

Q_B \ Q_A	0	1
0	0	1
1	X	X

$J_B = Q_A$

K_B

Q_B \ Q_A	0	1
0	X	X
1	0	1

$K_B = Q_A$

J_A

Q_B \ Q_A	0	1
0	1	X
1	1	X

$J_A = 1$

K_A

Q_B \ Q_A	0	1
0	X	1
1	X	1

$K_A = 1$

(e) Logic diagram :-

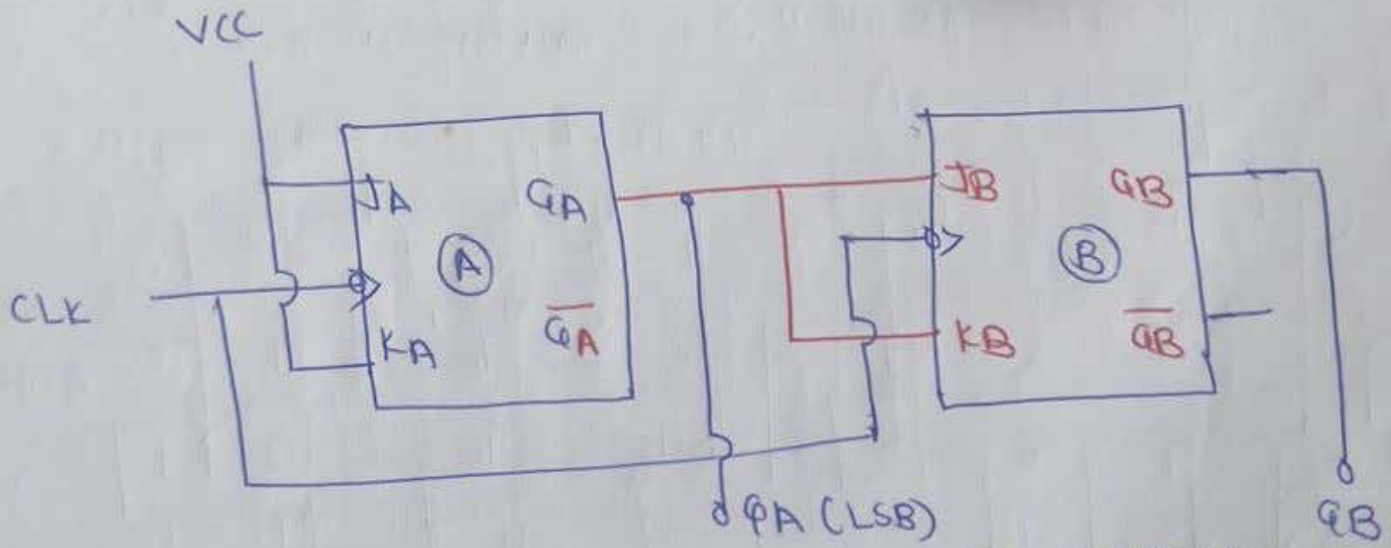


Fig:- 2-bit synchronous up counter.

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II) 3-bit synchronous up counter :-

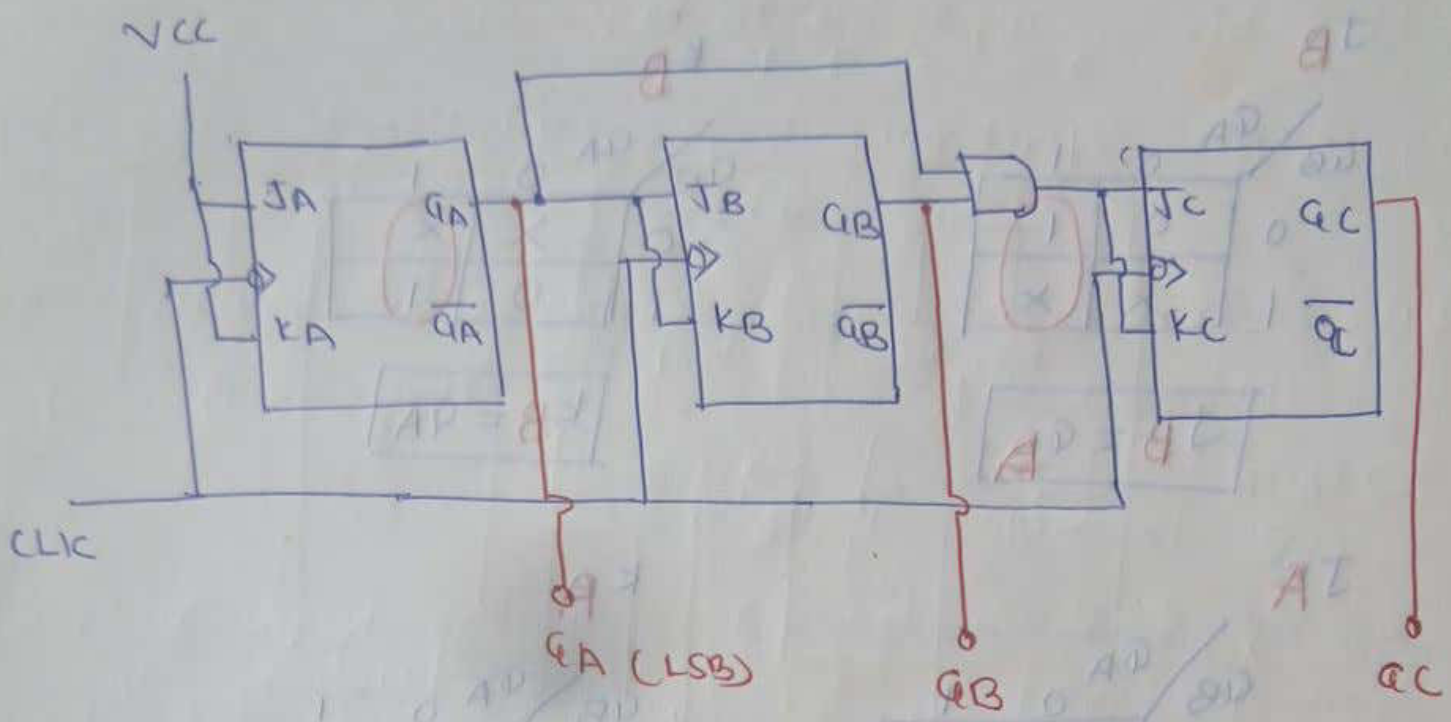


Fig :- 3-bit synchronous up counter.

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comparision of synchronous & Asynchronous counters -

Sr No.	parameter	Asynchronous counter	synchronous counter
1)	Logic ckt.	simple	Logic ckt. complicated
2)	clock i/p.	All the F/F are not clocked simultaneously	All the F/F clocked simultaneously.
3)	propagation delay	high P.D = n x (t_d) n = no. of F/F	low as compared to Asynchronous P.D = $(t_d)_{FF} + (t_d)_{gate}$
4)	Max^m freqn of operation	Low due to high P.D.	High due to shorter P.D.
5)	connection pattern.	O/p of preceding F/F is connected to clock of the next F/F.	There is no connection betn O/p of preceding F/F & CLK to next one.

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